# Late Medieval Unrest in the Holy Roman Empire

## Climate, Institutions, and Documented Unrest

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**Abstract**

I construct two linked datasets for the Holy Roman Empire, 1300–1520: a geolocated catalog of popular unrest and an annual panel of city-council representation, matched at city-year resolution across 306 cities. I ask whether unrest rises with wet-dry severity and whether it rises in the years institutions changed. Contrary to prior work's positive drought association, documented unrest tends to be more frequent in wetter years, and organized actors carry the pattern. The divergence from the closest published study survives changes in estimator, functional form, and event definition; sample scope and a recording artifact remain open. Guild representation's apparent associations with unrest come from the years in which representation changed, often as unrest's outcome; where guilds already held seats, severe drought went with less unrest. Burgher representation shows no association but precedes guild gains. Complaints are mostly political and fiscal. Every association here is descriptive.

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# 1 Introduction

Uprisings recurred throughout late medieval society, yet why they broke out where and when they did is only partly understood. Popular unrest in the Late Medieval Holy Roman Empire is well documented at the level of individual cities and regions but poorly documented at the level of the Empire as a whole. Dozens of well-archived cities carry detailed political histories individually (Bader & Dilcher, 1999; Isenmann, 2014).[1] Above the city, the record thins: pan-European conflict catalogs (Brecke, 1999; Chambru & Maneuvrier-Hervieu, 2024, p. 1088) cover the Empire only in passing. The institutional record has the same gap. Four panels code institutions for cities in the Empire (Bosker et al., 2013; Bosshart & Dittmar, 2026; Schaff, 2024; Wahl, 2016, 2019), but none records the unrest that so often accompanied institutional change.[2] Because they lack the context of contemporary social and political movements, causal tests and timing questions remain limited. As a result, debates over what drove unrest and how institutions shaped it remain unresolved for want of evidence at the right resolution. This paper presents two new datasets on unrest events and local institutional quality for cities in the Holy Roman Empire.

The first dataset is a geolocated, event-level catalog of popular unrest from 1300 to 1520 CE, coded by location, timing, participants, apparent grievance, and outcome. Most analyses of medieval unrest rest on a few well-known episodes or treat revolt as a fixed city-level attribute over long periods. Existing discussion of the circumstances of unrest focuses narrowly on particular regions or settings (e.g. Kaser (1902) discusses urban social movements in northern Germany, Franz (1977) and Blickle (1979) study central and southern peasant resistance, Hergemöller (2012) covers inner-city unrest, and so on). The lack of a comprehensive analysis leaves basic questions open. Did revolts cluster in particular institutional environments? Were they more likely in years of climatic volatility or plague? Did guild inclusion in city government precede unrest or follow it? The two datasets introduced here are built to put these questions to the test.

The second dataset is an annual city-level panel of political representation which tracks guild seats and broader burgher voice on city councils, built from the regional volumes of the *Deutsches Städtebuch*, a city-level encyclopedia of German urban history (Keyser, 1939, 1941, 1952, 1954, 1956, 1957, 1959, 1962, 1964; Keyser et al., 1995; Keyser et al., 2000; Keyser et al., 2003; Keyser & Stoob, 1971, 1974; Neitmann et al.,

[1] Non-exhaustively: Aachen (Meuthen, 1963); Andernach (Tille, 1901); Augsburg (Adrian, 2017; Frensdorff, 1865, p. XXXIV; Rogge, 1996); Basel (Heusler, 1860, p. 194 ff.); Bern (de Capitani, 1982); Braunschweig (Dürre, 1861, p. 151 ff.; Garzmann, 1986; Rotz, 1973, p. 209); Bremen (Schwarzwälder, 1975); Cologne (Bader & Dilcher, 1999, p. 565; Herborn, 1977, 1980; Looz-Corswarem, 1980; Maschke, 1980a, p. 226; Militzer, 1980); Erfurt (Mägdefrau, 1976); Freiburg im Breisgau (Maurer, 1891); Görlitz (Jecht, 1908); Halberstadt (Militzer & Przybilla, 1980); Hamburg (Postel, 1980; Stoob, 1980); Konstanz (Bechtold, 1981; Beyerle, 1898); Lübeck (Bader & Dilcher, 1999, p. 567; Maschke, 1980a, pp. 182–183; Rotz, 1977; Wehrmann, 1879); Magdeburg (Uitz, 1976); Mainz (Hegel, 1882, p. 72 ff.; Fischer, 1958); Marburg (Verscharen, 1985); Osnabrück (Haase, 1952); Regensburg (Ritscher, 1974); Speyer (Maschke, 1980a, pp. 123–124; Voltmer, 1981); Strasbourg (Alioth, 1988; Schmoller, 1875); Ulm (Jäger, 1831; Specker, 1997); Wrocław (Goerlitz, 1962); Würzburg (Willoweit, 2003); Zürich (Bluntschli, 1838; Largiadèr, 1936).

[2] Bosshart and Dittmar's conflict data, for instance, cover cities' disputes with their lords rather than unrest within them (Bosshart & Dittmar, 2026, p. 16).

2022). Three existing panels are coarser in time or coverage: they code council and guild representation only by century (Wahl, 2016, 2019); local participative government by century and only for cities that reached 10,000 inhabitants (Bosker et al., 2013); or council elections for only 35 cities at 25-year intervals from 1500 onward (Schaff, 2024). A fourth dates councils, mayors, charters, and the rules for selecting councils by year for more than 2,000 cities from 1200 to 1700, but has not yet been released (Bosshart & Dittmar, 2026).

To show what these data make possible, I link the two datasets to a common set of 306 cities, those with a known population above 1,000 by or before 1500 that lie roughly within the German-speaking part of the Empire (Bairoch et al., 1988; Buringh, 2021). I fit an illustrative regression of unrest on wet-dry severity, institutional representation, markets, population, and plague exposure on an annual panel of cities, and then benchmark the results against related work. One such study finds that severe drought raises the probability of conflict in European cities and, within the Empire, that the effect is muted where city councils are elected (Wigton-Jones, 2025). The drought pattern does not reproduce on my narrower 1300–1520 CE panel, and the institutional one only in part: burgher representation mutes severe-drought periods in five-year bins but not in annual data (Section 5). Drought never goes with more documented unrest at conventional significance, and in my own regressions wetter years go with more unrest, though in the main specification only at the 10% level (Section 4). The wet-year association belongs to organized actors: rebellions, and unrest led by guilds or elites. The online appendix shows that no single form of unrest carries a result that survives correction for the number of forms examined. I then rule out estimator choice and nonlinearity as explanations for the reversal. Geographic and temporal scope cannot be ruled out: on Wigton-Jones' own data, restricted to my cities and window, the drought coefficient for the outbreak of conflict is too imprecise to tell his effect from none. Nor does event definition explain it: I include peaceful episodes that fall outside the casualty threshold employed by Wigton-Jones (2025), but the violent episodes both datasets would count show the same drought-year deficit. Whether thin documentation of drought-year unrest contributes remains open (online appendix).

A further use of these data concerns local representation. Most work on institutions asks how they shape the activity of those operating within them; far less is known about what changes them (Callen et al., 2024). Representation and unrest are linked: representation can be the concession unrest extracts (Acemoglu & Robinson, 2000; Acemoglu & Robinson, 2005; Aidt & Franck, 2015; Aidt & Leon, 2016)—as it often was in late medieval cities (Bader & Dilcher, 1999; Dumolyn & Haemers, 2005; Rotz, 1985); it can be a channel that absorbs grievances before they become revolts (Blickle, 2000b; Lantschner, 2015; Ogilvie, 2019; Wigton-Jones, 2025); the corporate bodies through which it runs, guilds above all, can also organize violence (Blickle, 2000a; Lichbach, 1998; Planitz, 1965; Rotz, 1985); and the groups that win representation can close access behind them, turning a dispute channel into a source of grievance (North et al., 2009; Ogilvie, 2019; Scribner,

1994).

My results show that unrest clusters in the years in which guilds gained representation, often as its outcome, and those years produce an apparent guild association in either direction. Away from them, representation shows no clear association with unrest, within cities or between them. What differs with it is how a city reacted to crises: where guilds already held seats, severe-drought years were quieter. Burgher representation shows no association with unrest. Cities with burgher representation were more likely to give guilds voting seats later.

One caveat conditions every pattern in the paper. Larger and more politically prominent cities are more likely to have recorded institutions, population estimates, unrest episodes, and plague mortality. City fixed effects absorb only the part of that tendency that does not change over time, and no further control removes the rest.

The paper proceeds as follows. Section 2 describes how late medieval city councils were composed and contested, and how the council constitution entered unrest as its object, vehicle, and channel. Section 3 builds the two datasets and the city backbone that links them. Section 4 relates unrest to climate shocks and sets the result against Wigton-Jones's drought finding; Section 5 does the same for representation, with Wahl's panel as the comparison. Section 6 discusses what the new datasets imply for how the climate-conflict and institutions-conflict literatures treat pre-modern popular unrest in Central Europe.

## 2 Unrest and Institutional Change

Late medieval popular unrest in the Holy Roman Empire was distributed unevenly in ways that resist simple explanation. In the catalog, a year of severe drought or unusual wetness brings unrest to at most a few of the many cities it touches, and nearly all of their equally affected neighbors stay quiet. Accounts of revolt usually begin with grievance (how hardship becomes a complaint) and then ask how a complaint becomes collective action. In the cities this paper studies, the council constitution entered that sequence three times. It was the object of unrest, the arrangement much urban unrest set out to change. Its corporate bodies, the guilds above all, were a vehicle that made collective action possible. And where the community had a recognized voice, the constitution was a channel that could absorb a grievance before it became a revolt.

Late medieval cities in the Empire generally governed themselves through councils (*Rat*). The council was the organ of a legally recognized community (*commune*, *Gemeinde*) whose partial autonomy, with officeholders the burghers had some part in appointing, set the self-governing European city apart (Weber, 1978, p. 1226), but how officeholders were chosen and which corporate bodies held authority (*Stadtregiment*) differed from city to city and changed over time (Isenmann, 2014, pp. 243–244). Most cities were nominally

subject to a territorial lord, a bishop, or, for the free and imperial cities, the emperor directly, but day-to-day governance sat with the council (Isenmann, 2014, pp. 327–328).

The political arrangement inherited from the thirteenth century was primarily patrician. Councillors came from a circle of long-distance merchants, knightly families (*ministeriales*), or landowners (Maschke, 1980b, p. 171; Planitz, 1965, pp. 261–263). In the vast majority of urban councils, craftsmen rarely entered into government, and from the end of the thirteenth century the sitting council commonly filled its own vacancies by co-optation (Bader & Dilcher, 1999, p. 549; Isenmann, 2014, p. 355). Over the fourteenth and fifteenth centuries, many cities renegotiated that arrangement—craft guilds (*Zünfte*) won seats on the council, in some cities under a full guild constitution (*Zunftverfassung*), often as the settlement of an open conflict, and in several cities the old families later took the seats back (Bader & Dilcher, 1999, pp. 511–512; Isenmann, 2014, p. 234; Maschke, 1980b, pp. 170–171). Cities also varied in how far the wider body of resident burghers held any voice, from complete exclusion, through a supervisory (*Bürgerausschuss*) or minority position in council, to a citizen-chosen majority (Isenmann, 2014, p. 373), which is the range the panel's 0–3 burgher-representation scale records (Section 3).

These councils governed through the long agrarian depression (*Agrarkrise*) of the fourteenth and fifteenth centuries. From the late fourteenth century grain prices fell while craft wages rose, squeezing farm incomes until peasants left for the towns (Abel, 2013, pp. 52–53, 91; Campbell, 2016, p. 357; Slicher van Bath, 1963, p. 138). By the end of the Middle Ages about a fourth of settlements had been abandoned (Abel, 2013, p. 81; Slicher van Bath, 1963, p. 163). In the countryside, the peasant revolts of the period followed the slump (Slicher van Bath, 1963, p. 189), and German historiography reads them as a struggle between lords and peasants over the increasingly scarce resources (Blickle, 1979, pp. 231–232; Blickle, 2012, p. 62). In the towns, the agricultural depression-induced movement favored wage earners and craftsmen (Abel, 2013, pp. 53, 59), and the guilds that pressed for council seats were associations that had grown economically stronger as the cities flourished (Maschke, 1980b, p. 171).

Bad climatic years were never one-off. Wet summers damaged the harvests of 1401 and 1402, crops failed again in 1408, 1416, and 1426, and 1438 brought famine to regions across Central Europe (Abel, 2013, p. 59). Weather shocks of this kind are what this paper measures, and accounts of how such a shock becomes a grievance differ in where they place the trigger. Relative-deprivation theory holds that revolt follows the gap between what people have come to expect and what they receive, so that sustained improvement abruptly reversed is more explosive than persistent misery (Davies, 1962; Gurr, 2010). The depression had set up both forms of the gap: for townspeople, decades of cheap grain raised expectations that a failed harvest then reversed, and peasants squeezed by low prices could measure themselves against the town wages that drew others off the land. Rebellion also competes with production for a household's time and resources, so a shock

that lowers income lowers what revolt costs, and a poorer government is weaker at suppressing it (Fearon & Laitin, 2003, pp. 75–76; Grossman, 1991). A bad harvest did not lower every income, however. The agrarian depression set the low level around which grain prices moved, but each year's harvest still varied enough to matter, and because food demand is generally inelastic, a shortfall raised prices more than in proportion while a glut pushed them lower still (Slicher van Bath, 1963, pp. 118–119). Against that depressed baseline, a bad year cost households that bought grain, while farmers with grain to sell could do better than in an ordinary year.

The moral-economy account places the trigger in the response to the shock. Unrest follows the violation of a normative framework that defines what authorities may legitimately demand (J. C. Scott, 1976; Thompson, 1971), and food riots are better explained by changing grain policy than by a simple sequence of shortage, hunger, and riot (L. A. Tilly, 1971). In German cities, the provisioning duty that the moral economy presumes fell on the council, whose grain policy (including price controls and export bans) was explicitly meant to forestall rioting (Isenmann, 2014, pp. 978–980). In Hamburg's grain crisis of 1483, merchants were shipping grain to Iceland amid rising prices, which incensed local residents and became the root of the unrest (Stoob, 1980, p. 365). However, the comparative record of urban unrset leans toward the political rather than climatic. In a sample of 1,112 revolts and social movements from Italy, France, and Flanders up to 1425, risings sparked by scarcity made up about 2 percent, and after the Black Death revolts nearly tripled in frequency and turned largely on politics (Cohn, 2006, pp. 53, 74, 228, 231). This is similarly true in Germany (Table 1).

Unrest in the Empire more often set out to change the council itself. It was carried by trades that had grown economically stronger, and its most general trigger was the patrician councils' refusal to account for the state of the city's finances (Maschke, 1980b, pp. 171–172). Urban financial issues arose from bad policy—particularly foriegn policy (Rotz, 1985). Rather than extracting and immiserating the poor, councils frequently drew on the "financially responsible citizenry", whose standing complaints were that taxes were too high and that the council had mismanaged the town's administration (Rotz, 1985, pp. 66, 68–69, 71). The movements sought a share of the council and were rarely revolutionary in aim. Those that were not put down generally ended in revised compromise council constitutions (Bader & Dilcher, 1999, pp. 556, 562–564). The settlement was narrower than its form suggests. The guild councillors these constitutions installed were usually chosen by the council itself through co-optation, were nearly as wealthy and well connected as the sitting councillors, and remained a minority on most councils (Rotz, 1985, p. 69). For this reason the panel scores craft seats on a co-opted council as no guild representation (Section 3), and even where guilds did choose their own councillors, the men they sent were wealthy masters, so a recorded transition may widen the ruling circle among the propertied rather than open it to ordinary craftsmen. Even a decisive share

of the council could leave its policy unchanged: in Augsburg, once the craftsmen entered the council, they continued the armaments policy of the patrician families they had displaced (Maschke, 1980b, p. 173).

The economic theory of democratization explains why a council would concede seats at all. Extending political rights commits an elite to future redistribution in a way that a promise made under the old institutions cannot, since the promise lapses once the threat has passed. A temporary threat calls forth institutional change, and the threat is most credible when the opportunity cost of revolt is low, as in recessions (Acemoglu & Robinson, 2001; Acemoglu & Robinson, 2000). Empirical work finds the same link from unrest to institutional change. The Swing riots moved English voters toward pro-reform candidates and helped give the reformers the Commons majority they needed for the Reform Act of 1832 (Aidt & Franck, 2015), and in sub-Saharan Africa drought-induced riots raised the probability of democratic change, often through concessions by incumbents (Aidt & Leon, 2016). In medieval Flanders, revolt recurred often enough to form a "revolutionary tradition" in collective consciousness (Dumolyn & Haemers, 2005, p. 371). Its aim was access to the polity and control of the city's finances, and craft guilds won permanent representation after 1302 and had it confirmed in 1360 (Dumolyn & Haemers, 2005, pp. 375–376, 391). Because representation was so often the outcome of unrest, it cannot be read only as a background condition that unrest occurred within. A quiet stretch in the unrest record is at most evidence that political order was being maintained; it says nothing about whether the arrangements maintaining it were equitable or if tensions simmered just below the surface, waiting for a trigger.

Whatever the grievance, it became unrest only if someone bore the cost of organizing it. No individual has an incentive to bear the risk of acting first when the gains from success are shared regardless of who bore it (Lichbach, 1998; Olson, 2003; C. Tilly, 1978, pp. 62–63), and the solutions to that problem run through existing communities and hierarchies and through selective incentives that only organizations can effectively supply (Lichbach, 1998, pp. 15, 20; Olson, 2003, p. 51). A shock that absorbs households' time and resources can make that organizing harder even as it lowers the cost of unrest.Late medieval cities had organizations ready to hand—guilds weren't only organized for economic purpose, but also militarily, which might help explain the reasons for their fourteenth-century gains (Planitz, 1965, pp. 325–326). Successful movements institutionalized the new balance of power through existing associations built for other purposes (Bader & Dilcher, 1999, p. 563). Organized bodies also shaped the form unrest took: where people acted through recognized corporations such as guilds, contention tended to be local and direct (C. Tilly, 2005, pp. 45, 142), though late medieval revolts were often citywide and, north of the Alps, allied with peasants and other towns (Cohn, 2019, pp. 389, 394–395, 406). In the Low Countries, the contemporary vocabulary of collective action describes organized artisan action such as the *takehan*, an alliance with a common fund that financed strikes (Dumolyn & Haemers, 2017, pp. 47–48). Rotz (1985) argues that the German uprisings

drew on the guilds without being guild affairs alone, as the guilds were an available institution that the movements drew on (Rotz, 1985, pp. 72, 84–85).

Mamluk Egypt and Syria show the converse. There, so long as many notables collaborated with the regime, the associations on which European communes rested could not form, and without the religious scholars effective resistance was impossible to organize (Lapidus, 1984, pp. 190–191); a deviant Nile flood mattered politically by strengthening the bargaining position of the chief judge, who could coordinate a revolt (Chaney, 2013). On this reading, a shock's effect on unrest may show most clearly among actors already organized to act, which Section 4 examines by separating unrest by who carried it out.

Where a community had a recognized voice, the same organizations could press a grievance without revolt. *Kommunalismus*, Blickle's (2000b, pp. 148, 177) term for a social order built on the self-governing town and village community, supplied the grounds for doing so: a levy or tax could be contested because it endangered household subsistence (*Hausnotdurft*) or did not serve the common benefit (*Gemeiner Nutzen*). The strong, corporately constituted communities that carried these norms also supplied the organizational base for unrest, which was largely absent where they had not formed (Blickle, 2000a, pp. 261–262). Organized communities cut both ways, but representation changes the form a grievance takes. Guilds voiced their members' discontent in councils where they were represented, and otherwise through lobbying, petitioning, and litigation, with violence an occasional resort (Dumolyn & Haemers, 2005, p. 379; Ogilvie, 2019, p. 38). Guild or burgher representation can thus substitute for unrest, absorbing a drought as misfortune within a legitimate order and turning a lord's new levy into a petition or a suit; that substitution is the prior Section 5 tests. In a Holy Roman Empire subsample, Wigton-Jones (2025) finds drought-induced conflict muted where city councils were elected but not where they rested on burgher or guild representation. The closest analog to his elected councils in this panel is broader burgher representation, and his burgher-or-guild result runs against the guild half of the prior (Section 5).

*Kommunalismus* has clear limits. Blickle (2000b) first framed the model for Upper Germany alone and stresses the regional limit, and Scribner (1994, pp. 200, 202–204) observes that the thesis rests chiefly on south-west German and Swiss evidence, risks a reading in which medieval communes anticipate democracy, and is undercut by case studies in which communes were readily absorbed by hierarchy and oligarchy. Panel evidence on urban construction and political autonomy reaches a similar verdict: across German-speaking Europe, the core institutional gains cities won after the Black Death were non-democratic and served urban oligarchies (Bosshart & Dittmar, 2026, p. 32).

The movements for changing the makeup of councils should not be interpreted as democratic, particularly for the guilds, who often closed the deliberative channel over time. Whether guilds solved market failures in quality, training, and innovation or were distributional coalitions serving their members at others' expense

is an old debate (Ogilvie, 2019), and the council seats this paper tracks are its political side. In German cities the guild struggles were directed not only against patrician government but also against merchants, Jews, poorer craftsmen, journeymen, and wage-workers, and the guilds used the representation they won to keep such groups out (Ogilvie, 2019, pp. 39–40). The logic is that of a limited-access order, in which a dominant coalition secures peace by sharing the rents that limited entry creates and incorporates newly powerful groups when the balance of violence shifts (North et al., 2009). Guilds strong enough to threaten the patriciate were incorporated and then defended their new rents by restricting entry, so the organization that once channeled grievances could become a source of them. Whether guild seats dampened unrest or fed it may depend on how long they had been held, a question of timing that century-level institutional coding cannot resolve and the annual panel is built to address (Section 3).

A further channel opened at the level of the Empire, just after the study window. The imperial recess of 1526 referred disputes between lords and their subjects to the territorial courts and the imperial chamber court, and Schulze describes the result as a tendential juridification of social conflict, which gave subjects a route for grievances that did not run through revolt, a route Section 6 returns to (Schulze, 1980, pp. 76–77, 141).

# 3 Data

This paper's contribution is two new datasets for the Holy Roman Empire, 1300–1520 CE: a catalog of popular unrest and an annual panel of city-council representation, linked through a common backbone of 306 cities. This section describes each dataset's sources, construction, size, and coverage, and then the backbone that links them (Section 3.3). Section 4 relates the linked panel to climate shocks and Section 5 to institutional change, each set against the closest published study, Wigton-Jones's drought result and Wahl's panel. The climate, population, and plague covariates, matched to the same backbone, are introduced in Section 4, where they are used.

## 3.1 Unrest

I have compiled a catalog of 493 unrest episodes in the German-speaking Empire, 1300–1520, built for quantitative analysis of where and when unrest occurred rather than as a narrative history of each episode. To my knowledge it is the most comprehensive compilation for this region and period. The closest existing catalogs, that of Brecke (1999) and the Historical Social Conflict Database (HiSCoD) of Chambru & Maneuvrier-Hervieu (2024), cover the region only thinly, the first because it records mainly large-scale violence such as war, the second because about 92 percent of its entries come from France and England, whose

existing datasets it was built on, and no comparable survey exists for pre-industrial Germany (Chambru & Maneuvrier-Hervieu, 2024, p. 1088).[3]

The dataset is compiled primarily from secondary regional and urban histories and from encyclopedia descriptions of late medieval German unrest. While I draw on a wide array of sources (detailed in the appendix), the most substantive are:

- The regional volumes of the *Deutsches Städtebuch* encyclopedia;
- Turnau (2010), *Städtische Unruhen (Spätmittelalter)*, a *Historisches Lexikon Bayerns* reference entry cataloguing episodes of disorder in late medieval Bavaria;
- The *Historisches Lexikon der Schweiz*, an online encyclopedia which supplies episodes for Swiss cities and rural districts;
- Wikipedia articles in several languages, cited for individual episodes;
- Hergemöller (2012), *Uplop – Seditio: Innerstädtische Unruhen des 14. und 15. Jahrhunderts im engeren Reichsgebiet*, an extensive description of urban movements and riots;
- Franz (1977), *Der deutsche Bauernkrieg*, a systematic survey of rural and some urban unrest in southern Germany and Switzerland, particularly around the lead-up to the Peasants' War of 1525, and expanded on in Blickle (1979);
- Kaser (1902), *Zur politischen und sozialen Bewegung im deutschen Bürgertum des XV. und XVI. Jahrhunderts*, an extensive enumeration of urban revolts in the fifteenth and early sixteenth centuries; and
- Blickle (2012), *Unruhen in der ständischen Gesellschaft 1300–1800*, a general history of unrest in the period.[4]

The German historiography has a wide vocabulary for these events. Riot (*Aufruhr*, *Auflauf*), uprising (*Aufstand*, *Empörung*), revolt (*Revolte*), discord (*Zwietracht*), dispute (*Streit*), conspiracy (*Verschwörung*), civic and guild struggle (*Bürgerkämpfe*, *Zunftkämpfe*), and many others name points on a continuum of social and political conflict, each describing a different degree of intensity or an implicit teleology (Blickle, 2012, p. 5; Isenmann, 2014, p. 254). The clearest case is the East German proto-bourgeois revolution (*frühbürgerliche Revolution*), which read Reformation and Peasants' War together as an early bourgeois revolution at the start of the transition from feudalism to capitalism, prepared by a revolutionary crisis between 1470 and 1517 (Laube et al., 1982, p. 302; Steinmetz, 2021). I follow Blickle (2012) and use the more inclusive label "unrest" (*Unruhen*). Terms like rebellion, riot, or social conflict arrive with an implicit theory attached—a

[3] Wigton-Jones's conflict measure, which is built from Brecke (1999), is compared with this catalog in Section 4.

[4] A further large source is Peter Bierbrauer's research report on peasant revolts in the Empire, in Blickle (1980), which Bosshart & Dittmar (2026, pp. 77–78) describe as the most comprehensive body of evidence on peasant revolts before 1500. I have not yet been able to consult it, so episodes known only from it are not in the catalog.

view of what the event *was* and between *whom*—and I want to avoid prejudging episodes that differ widely in context, participation, extremity, and effect. *Bürgerkämpfe* shows the problem: it implies that essentially only citizens took part, which the sources cannot show, since they say little about the lower strata without citizenship and guilds sometimes included members without it (Isenmann, 2014, p. 255); *Bürger* itself can carry Marxist connotations (as in *bourgeois*) in certain contexts, but not in others. Even the sources do not agree on what to call an episode or who stood behind it, and support could shift within a single episode: a Hanseatic chronicle recorded the disturbance at Minden in 1405 as a rising of the citizenry, though the sworn alliance behind it came from councillors and the Forty, not the broader community (Ehbrecht, 1980), and a 1493 conspiracy in Korbach, believed to involve 120 citizens, lost the support of several conspirators when they were asked to swear a general oath, whereas the arrest of the Jauer conspirators in 1516 brought out the guilds and the common people (Johanek, 1997). "Unrest" sidesteps that debate and can accommodate the whole range of socio-political activity.

The inclusion criterion is accordingly broad: any collective action, actual or attempted, that asserted grievances or attempted to disrupt existing authority structures. Episodes suppressed before open confrontation are included where there is evidence of coordinated planning or a repressive response, and noble revolts against liege lords or the emperor are excluded. Chronicles do not always make clear whether unrest, rather than an ordinary dispute or settlement, has occurred; where the threshold is ambiguous I follow the interpretation of the source authors rather than impose a criterion of my own. The catalog therefore records the episode and its form, not a verdict on how broad its base was. In fact, most events captured are highly localized. The catalog records burghers against a city council (Speyer 1349), peasants against an appointed bailiff (Weingarten 1432), and townspeople against their prince-abbot (Kempten 1363), to name a few, each contesting a specific grievance with the immediate authority rather than the wider order. Not all unrest was violent, and not all involved popular protest—some were elite-led coups, others nonviolent demonstrations, refusals to perform homage, or symbolic acts of defiance—and the participants rarely sought to overturn the feudal order, aiming instead at negotiated political or economic concessions. The exceptions are notable for how late in the era they emerge: the Niklashausen episode of 1476, the Bundschuh conspiracies of the upper Rhine (1493–1517), and the Poor Konrad revolt of 1514, all of them eclipsed by the Peasants' War of 1524–25, which falls just after the study period. I end the window in 1520 because the central puzzle of that war, the formation of supraregional, cross-class alliances that turned rebellion into war, is a different question from what explains the scattered local uprisings of the preceding century (T. Scott, 1979, p. 953).

The source compilation records 722 rows in the study region and window, one for each place a revolt is recorded. Because a single revolt can unfold in many places, I treat the *episode*, not the raw row, as the primary unit of analysis: rows sharing a common originating-revolt identifier collapse into one episode, with

onset year taken as the earliest reported year among constituent rows and grievance flags aggregated by an any-flag rule. This yields 493 episodes. Only 5.1% of them span more than one place, but those account for 35.2% of the rows, so counting rows would let a handful of large revolts dominate the record. 31.6% of episodes are recorded as lasting more than one year (where no source reports an end year, it is set equal to the onset year).

Each episode is coded for one or more grievance types, where the source description names the grievance as a cause, demand, or trigger:

- Representation, or who sits in government—council seats, elections, eligibility for office, oversight committees
- Autonomy, or who rules the place and with what rights—jurisdiction, liberties against a lord, foreign rule, serfdom and tenure
- Administration, or how the authority governs—regulation of trade and crafts, the conduct of officials, failure to protect
- Fiscal, or the extraction burden imposed by the opposed authority—taxes, levies, dues, tithes, public debt
- Economic, or hardship from conditions beyond that authority's control—prices, wages, trade, and natural shocks such as drought, flood, harvest failure, epidemic, or fire
- Religious, or belief, worship, or church authority

The online appendix sets out the full definitions and the rules for boundary cases. Table 1 gives the resulting grievance mix by period, for the study region and the city-matched subsample. Disputes over who sat in government are the most common grievance early in the window, and fiscal grievance grows from 19.6% of episodes in the first half-century to 39.5% in the last, alongside a rising rural share. Economic grievance, which includes every named natural shock, by contrast stays marginal in every period (6.1% of episodes overall). Few revolts name a climate shock or hunger as a cause, which implies that subsistence stress rarely appears in the sources' account of why people engaged in unrest. Either it acted through the political and fiscal complaints the sources record, it went unrecorded, or it mattered little. How far this bears on the climate-conflict results is taken up in Section 4.

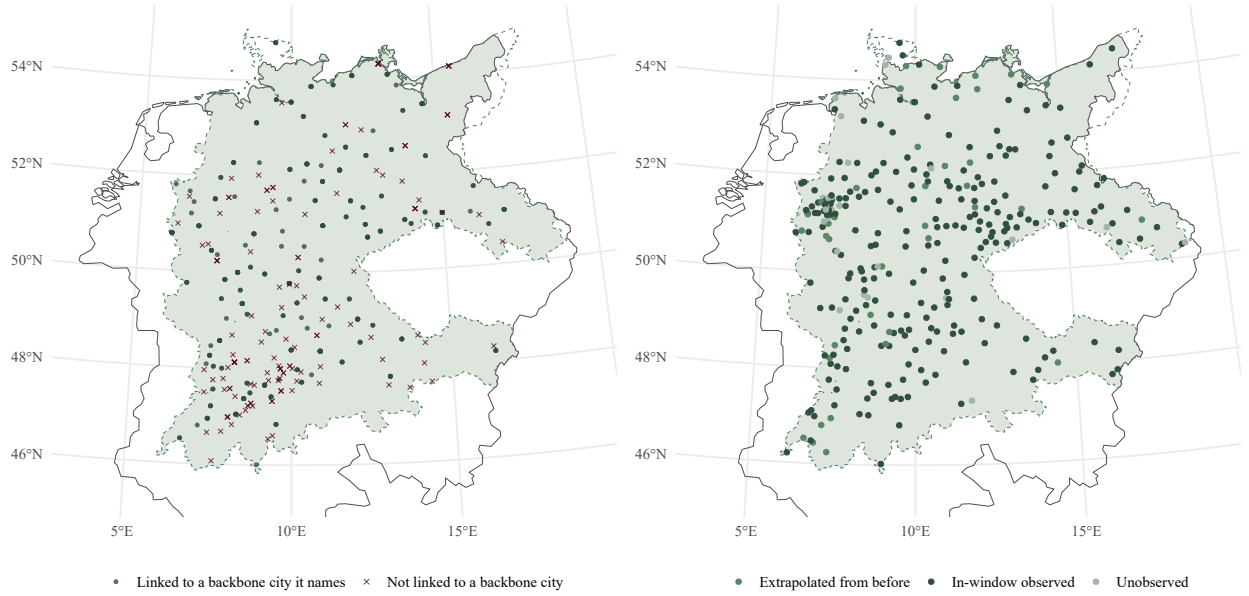
Each episode also carries a type, coded from the source text by what the participants attempted and how. The catalog uses six types: revolution (violent action aimed at a major change of the social or constitutional order, such as the Bundschuh conspiracies of 1502 and 1517), rebellion (open collective action by a large group, including guild or commune risings that depose or enter the council and risings against a lord), coup (a seizure of power, or an attempt at one, by a small group such as a faction or a few families), riot (short-

Table 1: Unrest episodes by period: grievance mix, setting, type, and violence, 1300–1520

| | 1300–49 | | 1350–99 | | 1400–49 | | 1450–99 | | 1500–20 | | All | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | SR | CM | SR | CM | SR | CM | SR | CM | SR | CM | SR | CM |
| *Episodes* | | | | | | | | | | | | |
| Episodes (N) | 97 | 80 | 105 | 76 | 115 | 84 | 100 | 61 | 76 | 48 | 493 | 349 |
| *Grievance type (percent of episodes)* | | | | | | | | | | | | |
| Representation | 42.3 | 46.2 | 33.3 | 38.2 | 36.5 | 50.0 | 24.0 | 29.5 | 25.0 | 27.1 | 32.7 | 39.8 |
| Autonomy | 20.6 | 17.5 | 20.0 | 18.4 | 24.3 | 15.5 | 28.0 | 24.6 | 19.7 | 16.7 | 22.7 | 18.3 |
| Administration | 10.3 | 11.2 | 20.0 | 22.4 | 19.1 | 20.2 | 24.0 | 29.5 | 23.7 | 20.8 | 19.3 | 20.3 |
| Fiscal | 19.6 | 20.0 | 28.6 | 28.9 | 32.2 | 36.9 | 37.0 | 44.3 | 39.5 | 41.7 | 31.0 | 33.2 |
| Economic | 7.2 | 8.8 | 5.7 | 6.6 | 1.7 | 1.2 | 8.0 | 9.8 | 9.2 | 14.6 | 6.1 | 7.4 |
| Religious | 4.1 | 5.0 | — | — | 3.5 | 3.6 | 8.0 | 11.5 | 2.6 | 4.2 | 3.7 | 4.6 |
| None | 25.8 | 23.8 | 33.3 | 30.3 | 25.2 | 22.6 | 26.0 | 21.3 | 36.8 | 41.7 | 29.0 | 26.9 |
| *Setting (percent of episodes)* | | | | | | | | | | | | |
| Urban | 89.7 | 96.2 | 91.4 | 98.7 | 83.5 | 95.2 | 69.0 | 88.5 | 75.0 | 87.5 | 82.2 | 94.0 |
| Rural | 5.2 | 1.2 | 7.6 | — | 20.0 | 10.7 | 30.0 | 14.8 | 32.9 | 20.8 | 18.5 | 8.3 |
| *Type (percent of episodes)* | | | | | | | | | | | | |
| Revolution | — | — | — | — | — | — | — | — | 2.6 | 4.2 | 0.4 | 0.6 |
| Rebellion | 46.4 | 47.5 | 44.8 | 46.1 | 42.6 | 47.6 | 40.0 | 45.9 | 32.9 | 35.4 | 41.8 | 45.3 |
| Coup | 11.3 | 13.8 | 7.6 | 5.3 | 4.3 | 4.8 | 9.0 | 13.1 | 1.3 | 2.1 | 6.9 | 8.0 |
| Riot | 7.2 | 8.8 | 6.7 | 9.2 | 11.3 | 13.1 | 5.0 | 3.3 | 6.6 | 8.3 | 7.5 | 8.9 |
| Protest | 17.5 | 13.8 | 21.0 | 19.7 | 32.2 | 28.6 | 26.0 | 27.9 | 19.7 | 12.5 | 23.7 | 20.9 |
| Unclear | 10.3 | 8.8 | 12.4 | 10.5 | 17.4 | 15.5 | 21.0 | 18.0 | 23.7 | 18.8 | 16.6 | 13.8 |
| Not recorded | 12.4 | 12.5 | 12.4 | 14.5 | 4.3 | 3.6 | 6.0 | 3.3 | 13.2 | 18.8 | 9.3 | 10.0 |
| *Violence tier (percent of episodes)* | | | | | | | | | | | | |
| Violent | 61.9 | 66.2 | 58.1 | 59.2 | 54.8 | 60.7 | 52.0 | 59.0 | 43.4 | 50.0 | 54.6 | 59.9 |
| Peaceful | 15.5 | 12.5 | 17.1 | 15.8 | 23.5 | 20.2 | 21.0 | 19.7 | 19.7 | 12.5 | 19.5 | 16.3 |
| Ambiguous | 10.3 | 8.8 | 12.4 | 10.5 | 17.4 | 15.5 | 21.0 | 18.0 | 23.7 | 18.8 | 16.6 | 13.8 |
| Unknown | 12.4 | 12.5 | 12.4 | 14.5 | 4.3 | 3.6 | 6.0 | 3.3 | 13.2 | 18.8 | 9.3 | 10.0 |

*Notes:*
SR = study region: the 493 episodes geocoded inside the German-speaking region, plus non-geocoded episodes the source places in Germany. CM = city-matched: the 349 episodes linked to a study-population city. Grievance types, unrest types, and the urban and rural flags are non-exclusive, so they can sum to more than 100; None is the share with no grievance flag. Violence rows sum to 100. A dash marks exactly zero.

lived crowd violence without organization), protest (nonviolent collective action, including petitions, refusals of dues or homage, and strikes), and unclear, where the sources do not specify what form the unrest took. Pogroms with no other aims are catalogued separately and do not count as unrest here (an episode that also turned on Jewish residents keeps its type and records them as a target), and conflicts between authorities, such as feuds and wars, are excluded. An episode takes every type its source rows record, so one whose sources describe both a protest and a rebellion counts under both. Table 1 reports the types by period. Rebellion is the most common, at 41.8% of episodes, and revolution in the strict sense appears only in the last two decades of the window. The type field is empty for 6.4% of source rows, and I leave those episodes as not recorded rather than impute a type. Two coarser codings derive from the type. Severity is major for revolution, rebellion, and coup, minor for riot and protest, and unknown otherwise: 48.7%, 25.4%, and 26.0% of episodes. The violence tier, also in Table 1, is violent for revolution, rebellion, coup, and riot,




(a) Unrest episodes, linked and not linked

(b) Institutional coverage of study-population cities

Figure 1: Unrest episodes and institutional coverage in the study region, 1300–1520. The solid gray line is the Holy Roman Empire's c.1378 border and the dashed green line the edge of the German-speaking region; their shaded intersection is the study region. Panel (a): geocoded unrest episodes, linked to a backbone city whose name they bear or not linked; a linked episode appears at each linked place. Panel (b): the 306 study-population cities by institutional record: entries inside the window (78.8%), entries only before 1300, carried forward (12.7%), or none (8.5%).

peaceful for protest, and ambiguous for unclear. An episode's type is a coding of how the sources describe it, not an independent measurement.

Two source-level concerns condition every pattern drawn from this catalog. The first is the ordinary survivorship problem: no systematic treatment of medieval revolt is ever complete, and even for intensively studied regions, archival work on unrelated topics keeps surfacing episodes absent from the canonical secondary literature (Blickle, 1979). The second is novelty: this is the first large-scale, georeferenced catalog of its kind, so there is no earlier compilation of comparable scope or resolution against which to benchmark its coverage. The catalog should therefore be read as a lower bound on unrest incidence, weighted toward episodes prominent enough to draw chronicle notice or secondary commentary. Because each underlying source has its own regional, thematic, or temporal focus, the quality of the documentation may also vary across region and time. Differences in the number of recorded episodes across regions should therefore not be read automatically as differences in incidence without corroboration. A thinner record may reflect fewer episodes or it may be thinner coverage, and these data alone cannot distinguish the two.

## 3.2 Institutions

The institutional dataset is an annual panel of city constitutions, built from a record of changes, descriptions, impositions, and reforms—one row per documented entry, not an annual observation—that holds 1,010 dated changes in 306 of the 306 cities of the common backbone (Section 3.3), the same cities to which the unrest catalog and the other sources are matched. It codes three variables: guild representation and burgher representation, each on an ordinal 0–3 scale, and imperial status; 0/1 versions of the two scales, derived from them, are used in the main tests and the comparisons described below. It is more detailed than the other institutional panels available for the Empire. The Participative Political Institutions Database (Wahl, 2016, 2019) codes representation at century marks, Bosker et al. (2013) code local participative government by century and only for cities that reached 10,000 inhabitants, and the city-level data in Schaff (2024) record whether a city held council elections at 25-year intervals, rather than annually, coarsening the same kind of transition record that this paper converts directly into an annual series. The city-institution and political-fragmentation panel in Bosshart & Dittmar (2026, p. 16) is closer in resolution: it dates its variables directly at the city level, typically by year, and likewise draws its council, charter, and collective-action variables directly from the *Städtebuch*. It is not publicly released at the time of writing, however, so no direct comparison of the two panels' coded values is possible yet.

The constitutional record is derived primarily from three encyclopedic sources. For German cities it draws on the *Deutsches Städtebuch*'s regional volumes (Keyser, 1939, 1941, 1952, 1954, 1956, 1957, 1959, 1962, 1964; Keyser et al., 1995; Keyser et al., 2000; Keyser et al., 2003; Keyser & Stoob, 1971, 1974; Neitmann et al., 2022), for Austrian cities it draws on the *Österreichisches Städtebuch* (Baltzarek et al., 1973; Csendes et al., 1999; Goldmann et al., 1976; Goldmann et al., 1982; Goldmann et al., 1988; Hoffmann, 1968; Hye et al., 1980; Hye, 2001; Knittler, 1970), and for Swiss cities on the *Historisches Lexikon der Schweiz* (HLS). Supplementary material, chiefly an Alsatian historical dictionary and Wikipedia, fills in any gaps.

Guild representation measures how far guilds took part in governing the city, and burgher representation measures how far the citizenry as a whole chose its government. The two are scored separately, so that guild delegates count only toward the first. Each runs from 0 to 3, and Table 2 gives the criteria for every score, the treatment of compulsory guild-type electoral units, and the derived indicator for guild seats. A second derived indicator, burgher seats, is 1 where burgher representation is 2 or higher and 0 otherwise. The third variable, imperial status, is taken from the territorial histories of *Princes and Townspeople* (Cantoni et al., 2019) and cross-checked against the *Städtebuch* and Wikipedia; a loss of status is recorded as an entry in the log.

The tests use these variables in different forms. The regressions of Section 4 and Section 5 enter guild

Table 2: Coding of the institutional variables

| Variable | Score | Criteria |
|---|---|---|
| Guild representation (0–3) | 0 | No formal guild role, including a stated absence, abolition by the lord, craft seats reserved on a council filled by co-optation or by the lord, and a merchant guild that is itself the co-opting ruling elite. |
| | 1 | An advisory or consultative channel only, or guilds that elect, or dominate the college that elects, the council without holding council seats. |
| | 2 | Guild representatives hold voting seats on the council, half or fewer of its members (exactly half scores 2). |
| | 3 | A guild majority on the council, or a formal guild constitution (*Zunftverfassung*). |
| Burgher representation (0–3) | 0 | Rule by a lord or retainers, or a hereditary mayor. |
| | 1 | A patrician or co-opted council, including one elected by the council itself. |
| | 2 | Limited representation: a citizen committee beside a co-opted council, a single delegate, or half or fewer of the council chosen by the citizenry. |
| | 3 | A majority of the governing council, or of the executive, chosen by the citizenry. |
| Guild seats (0/1) | 0 | Guild representation of 0 or 1: guilds hold no voting seats on the council. |
| | 1 | Guild representation of 2 or 3: guild representatives hold voting seats on the council. |
| Burgher seats (0/1) | 0 | Burgher representation of 0 or 1: a lord, his retainers, or a patrician or co-opted council governs. |
| | 1 | Burgher representation of 2 or 3: the citizenry chooses at least part of the council or its executive. |
| Imperial status (0/1) | 0 | A city under a territorial lord, including a city whose imperial standing was claimed but never conceded or was pledged away and not redeemed. |
| | 1 | A free, imperial, or free imperial city, immediate to the Empire, from the year the status is documented until it is lost, for example by mediatization. |

*Notes:*
Scores are assigned to each documented entry and carried forward to the next; a cell the sources cannot settle inherits the previous value. Compulsory guild-type electoral units covering the whole citizenry (the Cologne Gaffeln, the Strasbourg tribus) score as guild representation by their share of council seats (3 for a majority, 2 for half or fewer), and election through them counts as election by the citizenry on the same rule. Guild delegates do not count toward burgher representation.

seats and burgher seats on the annual panel, in the same year and lagged one year, and, as a robustness check, as the share of each five-year period in which they are held; the 0–3 scales and imperial status enter only as robustness variants. The comparisons with Wahl and Wigton-Jones also rely on the 0/1 versions. The Wahl cross-walk below compares guild seats and burgher seats with Wahl's coded guild participation and burgher representation. The guild-revolt replication takes a gain of guild seats as its outcome. The Wigton-Jones benchmark builds its three-tier government type from the guild-seats and burgher-seats indicators.

Each entry is scored from the source text against the rules set out in Table 2. I then convert the resulting record to an annual panel by carrying values forward for each city under the assumption that a constitutional change persists until the record states otherwise. This assumes no unobserved reversal between recorded changes, and mirrors the convention in other institutional panels (Schaff, 2024; Wahl, 2016, 2019). The scores

still rest on judgment calls, most visibly the treatment of reserved seats, electoral colleges, and merchant elites, and a sensitivity check across those rulings is future work. Where the source does not settle a cell, I leave it blank and the cell inherits the previous entry's value. The years before a city's first recorded entry are left missing rather than backward-filled, because the early record is thin.[5] Governance before the Late Middle Ages (except for a few large cities) is not well recorded. Before about 1200, towns were governed mainly through the town lord's officials, such as a *Vogt* or *Schultheiß*, working with citizen judges and advisers; councils first appear in the sources from the end of the 12th century and spread to some 150 towns by 1250 and about 400 by 1300 (Isenmann, 2014, pp. 216–217, 220; Planitz, 1965, p. 297). In this dataset, 169 of the 306 cities (55.2%) have a documented council by 1300, and 238 by 1400. At a city's first recorded entry, burgher representation is already nonzero in 68.1% of cities (188), while guild representation is absent in 95.4% (251). This is consistent with the record usually beginning once a council is already in place, and with guild participation arriving later as a reform, mostly in the fourteenth century: 49 of the 87 cities that reach guild seats do so between 1300 and 1399, with a median year of 1382. Over 1300–1520, the resulting annual panel covers 67,626 city-years across 306 of the 306 cities. By city, 78.8% have at least one documented entry inside the window, 12.7% rest on entries from before 1300 carried forward, and 8.5% have no entry in or before it. By city-year, counting all 306 cities, 83% carry a coded value, but only 55.4% rest directly on an in-window entry; another 27.8% are carried forward from an earlier one. Figure 1b shows the city-level classification geographically.

The new institutional panel shares much of its archival lineage with Wahl's panel, so its novelty lies in resolution and construction. I validate it against Wahl's panel at his century marks inside the window (1300, 1400, 1500), comparing his coded institutional state with burgher seats and guild seats. An annual panel built from a record of changes and a century-snapshot panel can legitimately disagree about a given city-year, so the question is whether they disagree at random or at a threshold. I decompose the disagreement two ways: by direction, which panel codes representation present when the other does not, and by coverage class, whether a city-year's value rests on a change documented inside the window or is carried forward from before 1300.

Of the 306 study-population cities, 222 (72.5%) appear in Wahl's database; the unmatched remainder reflects in part that his database covers Germany only and leaves out the Alsatian and Swiss cities in the study population (Section 3.3). At his century marks inside the window, the comparison covers 244 city-years in 135 cities. Agreement is moderate for burgher representation and higher for guild representation: the burgher-seats indicator matches Wahl's coded burgher representation in 66.7% of 234 pairs, and guild seats match his coded guild participation in 81.4% of 231. I read this as informative disagreement. The

[5] A sensitivity check on this choice for guild and burgher representation is in the online appendix.

panels differ in source lineage, in granularity, and plausibly in where they draw the line on what counts as institutionalized representation; near-perfect agreement would have raised the opposite concern, that the new panel merely re-derives the old one.

For guild representation the disagreement runs almost entirely one way: of the 43 guild disagreements, 38 are city-years this panel codes as represented and Wahl does not, against 5 the other way. For burgher representation the split is close to even, 43 against 35 of 78. The base rates say the same: on the compared city-years (231 for guild, 234 for burgher representation), Wahl codes guild participation present in 13.9% of city-years and this panel in 28.1%, while for burgher representation the two sit close together, 26.9% against 30.3%. Symmetric coding error cannot produce the guild asymmetry; a difference in threshold can. A panel that counts guild voting seats short of a full guild constitution will code guild representation present in city-years that a century-snapshot panel, keyed to a formal guild constitution, codes as absent. The two measure adjacent constructs.

Carrying values forward matters too, though not as one might expect. Agreement is *higher* where a city-year's value is carried forward from before 1300: 100.0% for guild seats (83 pairs) and 89.5% for burgher representation (86), against 70.9% and 53.4% where a change is documented inside the window. Carried-forward city-years mostly belong to cities whose arrangements were settled well before 1300, which both panels pick up the same way; the disagreement concentrates where the record documents new changes, which the two panels register differently. Two consequences follow for this paper. The burgher-representation measure is least certain around newly documented changes inside the window, which is also where the *Kommunalismus* prior draws its variation. The guild measure is internally consistent but sits systematically above Wahl's threshold, so the guild coefficients here are not directly comparable in level to results estimated on his coding. I treat neither panel as ground truth.

### 3.3 City backbone

The backbone is every city in the Buringh-lineage population dataset (Buringh, 2021) that lies in the German-speaking part of the Empire, within its c. 1378 boundary, and records 1,000 or more inhabitants in 1300, 1400, or 1500. The dataset lists every city that reached that size by 2000, so the population condition removes later foundations such as Karlsruhe and Wolfsburg. These cities mostly match with the *Princes and Townspeople* dataset (Bogucka et al., 2019), which adds helpful identifiers and linkage to their data (also drawn from the *Deutsches Städtebuch*), but is not a condition for inclusion. Three further entries are left out: Metz and Thionville, Lorraine cities of the Three Bishoprics that fall inside the language boundary but outside the Alsatian and German source volumes, and modern Wuppertal, whose constituent towns Barmen

and Elberfeld are already in the backbone. The population data share their lineage with the Participative Political Institutions database (Wahl, 2016, 2019) and the Wigton-Jones benchmark (Wigton-Jones, 2025) in Section 4, which keeps the backbone comparable to both. The institutional volumes for Alsace and Switzerland number cities differently from the backbone, so those cities are linked by name; 31 of the 33 link to the study population, and the other two are Metz and Thionville, which the source sheet had listed with Alsace.

# 4 Climate and Unrest

To illustrate what the combined data make possible, I ask whether documented unrest in the backbone cities rises or falls with climate shocks, and whether that association differs with a city's institutions. Weather shocks have been tied to conflict in many settings: colder growing seasons raised the persecution of Jews in pre-industrial Europe (Anderson et al., 2017), droughts and floods increased peasant revolts in imperial China (Jia, 2014), drought did the same in revolutionary France (Waldinger, 2024) and drove nomadic attacks on China (Bai & Kung, 2011), and negative rainfall-driven growth shocks raised the risk of civil conflict in modern Africa (Miguel et al., 2004). Institutions shape how such shocks play out. Nile floods shifted political power in Egypt toward religious authorities (Chaney, 2013), the Black Death pogroms in the Holy Roman Empire were less intense and violent where the emperor governed directly (Finley & Koyama, 2018), and in an Empire subsample drought-induced conflict was muted where city councils were elected (Wigton-Jones, 2025). This section takes up the climate question and the published result that bears on it; Section 5 takes up the institutional one.

The regressions bring the two new datasets together on the 306-city backbone. An episode is linked to a backbone city when one of the places it is recorded at lies within 7.7 km of the city and bears the city's name or a variant of it, or lies within 50 km and bears exactly the city's name. Villages and abbeys bearing another name do not count for a city, regions count only where the city ruled them, and an episode recorded in several backbone cities counts for each. Of the 493 study-region episodes, 349 link to a backbone city (Figure 1a); these city-matched episodes are the outcome in every regression that follows. The linked subsample is more urban than the full catalog (Table 1), so the results speak to urban unrest more than to rural. Wet-dry severity comes from a historical climate reconstruction of the self-calibrating Palmer Drought Severity Index (scPDSI) (Cook et al., 2015); negative values indicate drought, so a positive coefficient means wetter conditions. Population comes from Buringh (2021), which records towns of 1,000 or more at century marks; each year takes the city's most recent snapshot (1300, 1400, or 1500), with years before a city's first snapshot set to 500 and gaps between snapshots interpolated linearly. Plague exposure data come from a

geocoded dataset of European plague outbreaks (Schmid et al., 2015) supplemented with the plague entries of the *Deutsches Städtebuch*. Rye suitability is the potential yield of rain-fed winter rye under low inputs from the Global Agro-Ecological Zones (GAEZ) project of the Food and Agriculture Organization, the crop-yield data behind the Caloric Suitability Index, extracted at each backbone city's coordinates (Galor & Özak, 2015b, 2015a, 2016); it does not vary over time within a city, so it enters only through its interaction with wet-dry severity. Market rights come from the markets series of the *Princes and Townspeople* collection (Cantoni et al., 2020), which is supplemented for Swiss, Alsatian, and Austrian cities using the *Historisches Lexikon der Schweiz*, the *Dictionnaire historique des institutions de l'Alsace*, and the *Österreichisches Städtebuch*. A city counts as holding a market from its first documented grant or mention, and loses it when a recorded withdrawal cancels its grants.

The outcome is one when a city-matched unrest episode falls in a city in a given year and zero otherwise. The main specification is a linear probability model with wet-dry severity, guild and burgher representation, a market indicator, log population, and a plague indicator as covariates, and city and year fixed effects. The accounts in Section 2 give the climate shock no single predicted sign. Deprivation and opportunity-cost accounts predict more unrest after a bad year, the moral-economy account predicts it only where the response breaches customary entitlement, and the collective-action account predicts less, since a shock that absorbs households' time and resources makes organizing harder. Because those accounts speak to bad years, I also split the distribution into its tails, with indicators for a city's driest and wettest tenth of years against the middle 80 percent; a linear term alone cannot show whether an association sits in drought years, in wet years, or in both. Scarcity also cut both ways for councils: they fixed maximum grain prices, sold from communal stores below the market price, and banned exports in times of dearth to forestall unrest (Isenmann, 2014, pp. 978–980; see also Dameron, 2017),[6] but the cost could itself breed fiscal grievance, as when Basel borrowed the equivalent of a full year's budget in 1438/39 to buy grain and build a granary (Isenmann, 2014, p. 980).Log population and the plague indicator enter as the covariates most likely to drive both unrest and its survival in the record.

The models are estimated on the annual panel, the resolution of the climate-conflict literature and of the benchmark below. Unrest is rare at that resolution, recorded in 0.67% of city-years in the main sample, so I read every coefficient against that base rate. A logit or probit with city fixed effects would drop every city that never records unrest, since its outcome never varies (Timoneda, 2021).[7] The linear model keeps every city and handles the fixed effects directly. Standard errors allow for correlation across space and over time (Colella et al., 2019; Conley, 1999): between cities within 100 km of each other in the same year, and within

[6] Nuremberg did all three in 1437, under popular pressure (Isenmann, 2014, p. 980).

[7] Fixed-effects logit and probit estimators also suffer from the incidental-parameters problem, the bias that arises when many fixed effects are each estimated from few observations; it shrinks as the number of observations per city grows (Greene, 2004).

Table 3: Climate timing: wet-dry severity and its deciles, this year and last

| | Any unrest × 100 | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | Same-year climate | | | | Prior-year climate | | | |
| Model: | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
| *Variables* | | | | | | | | |
| scPDSI | 0.0620** | 0.0698 | | | 0.0505* | -0.0205 | | |
| | (0.0278) | (0.0705) | | | (0.0267) | (0.0626) | | |
| scPDSI × Rye suit. | | -0.0047 | | | | 0.0421 | | |
| | | (0.0385) | | | | (0.0355) | | |
| Bottom-decile scPDSI | | | -0.0874 | -0.1672 | | | -0.0310 | 0.2318 |
| | | | (0.1288) | (0.4233) | | | (0.1234) | (0.4223) |
| Top-decile scPDSI | | | 0.1930 | 0.7741 | | | 0.1612 | -0.8615 |
| | | | (0.1296) | (0.5385) | | | (0.1342) | (0.5957) |
| Bottom-decile scPDSI × Rye suit. | | | | 0.0485 | | | | -0.1591 |
| | | | | (0.2700) | | | | (0.2541) |
| Top-decile scPDSI × Rye suit. | | | | -0.3444 | | | | 0.6060 |
| | | | | (0.3161) | | | | (0.3724) |
| Log population | 0.0572 | 0.0572 | 0.0551 | 0.0547 | 0.0567 | 0.0567 | 0.0424 | 0.0432 |
| | (0.1028) | (0.1028) | (0.1030) | (0.1030) | (0.1029) | (0.1029) | (0.1032) | (0.1032) |
| *At median rye suit. (1.70 t/ha)* | | | | | | | | |
| scPDSI | | 0.0619** | | | | 0.0510* | | |
| | | (0.0278) | | | | (0.0268) | | |
| Bottom-decile scPDSI | | | | -0.0849 | | | | -0.0383 |
| | | | | (0.1327) | | | | (0.1251) |
| Top-decile scPDSI | | | | 0.1895 | | | | 0.1672 |
| | | | | (0.1298) | | | | (0.1356) |
| City | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Year | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| *Fit statistics* | | | | | | | | |
| Observations | 67,626 | 67,626 | 67,626 | 67,626 | 67,626 | 67,626 | 67,320 | 67,320 |
| $R^2$ | 0.02053 | 0.02053 | 0.02047 | 0.02048 | 0.02050 | 0.02050 | 0.02037 | 0.02039 |

*Conley standard errors (100 km, 50 years) in parentheses*
*Signif. codes: ***: 0.01, **: 0.05, *: 0.1*

Linear probability models on the annual city-year panel, 1300–1520, all 306 backbone cities; outcome 100 if unrest, 0 otherwise, so coefficients are in percentage points. Columns 1–4 use the same year's conditions and 5–8 the year before; within each, the first two enter scPDSI (negative in drought) and the last two its bottom and top deciles. Even columns add the rye interaction and report the climate terms at zero suitability; the rows below the line give them at the median city.

a city across years up to 50 years apart. The weights fall linearly with distance and with time, which keeps the estimated variance from turning negative (Newey & West, 1987). The online appendix reports equal weights, city-clustered standard errors, and other cutoffs.

Table 3 starts with climate alone, in the same year and the year before, and asks two further questions: whether the association depends on a city's agricultural endowment, through an interaction with rye suitability, and whether it holds at both ends of the distribution, through indicators for the driest and wettest tenth of city-years. The association is clearest for the same year's conditions. Across all 306 cities, including those without an institutional record, unrest breaks out in just 0.52% of city-years, and a year one standard deviation wetter than usual sees roughly 14% more of it (column 1). That standard deviation, 1.2 scPDSI units, is how far a year typically departs from its city's average and from that year's average across cities, so the comparison is with a city's own usual conditions. Because the preindustrial past was drier than the twentieth-century calibration period (Cook et al., 2015, p. 5), with an average city-year scPDSI

Table 4: Same-year climate and unrest by type and perpetrator

| Subset | Episodes | scPDSI (per SD) | Bottom decile | Top decile |
|---|---|---|---|---|
| All unrest | 349 | +14%** | −17% | +37% |
| *By type* | | | | |
| Rebellion | 158 | +23%** | +6% | +6% |
| Protest | 73 | +23% | −119%** | +45% |
| Unclear | 48 | +22% | −62% | +26% |
| Riot | 31 | −10% | −4% | +5% |
| Coup | 28 | +27% | −39% | +123% |
| *By perpetrator* | | | | |
| Guilds | 154 | +20%** | −42% | +48% |
| Burghers or commune | 171 | +4% | +21% | +16% |
| Elites | 60 | +20%** | −41% | +68% |
| Peasants | 27 | +40% | −36% | +28% |

*Notes:*
Each row repeats columns 1 and 3 of Table 3 with the outcome restricted to the episodes of one type or perpetrator group, and reports each coefficient as a percent of that subset's own base rate. The scPDSI column is the change for a year one within-city standard deviation wetter than usual; the decile columns compare a bottom- or top-decile year with a year in the middle 80 percent. Types and perpetrator groups overlap, since an episode counts for every one its sources name. Conley standard errors; *** 0.01, ** 0.05, * 0.1.

of -0.9 here, much of that variation runs from more drought to less. Neither tail carries the association on its own: top-decile years see about 37% more unrest and bottom-decile years about 17% less, but neither difference is significant in any column. The bottom decile (median −5.0) is severe drought, while the top decile (median 2.7) is mostly only moderately wet. Last year's climate points the same way but more weakly, and only its scPDSI terms reach even the 10% level. Rye suitability does not condition the association in either year: none of the interactions is significant, and at the median city's suitability a year one standard deviation wetter than usual still goes with about 14% more unrest. A wetter-than-usual year going with more unrest runs against the work in which drought or income loss raises conflict (Miguel et al., 2004) and against deprivation accounts that expect more unrest after a bad year (Gurr, 2010); it is the direction the collective-action account allows.

Table 4 splits the outcome by what the unrest was and by who carried it out. In a year one standard deviation wetter than usual, rebellions rise by about 23%, unrest led by guilds by about 20%, and unrest led by urban elites by about 20%, each against its own base rate and each significant. The association is not universal. Unrest led by burghers or the commune, the largest group, shows almost none (about 4%), and riots show none, so the wet-year association belongs to organized actors rather than to unrest in general; protest, coups, and peasant risings point the same way but rest on too few episodes to read. No group's top-decile rise is significant. Protest moves at the other end. In a bottom-decile year recorded protest all but disappears, the only subset whose drought tail is significant. The estimated drop exceeds protest's own

base rate, which a linear model allows and the data cannot, so I read it as a sign rather than a size. Both patterns fit the collective-action account. On this reading, guilds and elites are small, organized groups that can act at low cost and press their claims when conditions are good, while the petitions, refusals, and strikes that make up protest draw on time and resources that a severe drought takes away. The subsets overlap and several are small, so the consistency of the pattern carries more weight than any one coefficient. The association also keeps its sign on five-year periods and in a Poisson model of the unrest count, where it is significant (online appendix).

Wigton-Jones (2025) studies the drought-conflict relationship, moderated by local government type, across 673 European cities from 900 to 1800, and finds that more severe droughts raise the probability of conflict. On a 324-city Holy Roman Empire subsample built on Wahl's panel, the effect is muted where councils are elected and persists where representation runs through burghers or guilds. His conflict measure is built differently from my catalog, a difference the diagnosis below takes up. A single comparison on my own data could not tell a failure of his pattern from an error in my implementation, so I first fit an approximation on my own data, with my implementation checked on his, and then re-fit mine with his estimator. A diagnosis of the remaining divergence follows.

The approximate benchmark mirrors his functional form as closely as my variables allow, on five-year city-periods: a severe-drought indicator, the driest tenth of periods and so the bottom-decile cut used above, interacted with a three-tier government type built from guild and burgher representation, on the 254 backbone cities with a recorded constitution, the institutional sample that Section 5 uses throughout. Burgher/guild cities are those where guild representatives hold voting seats in at least half the period; broader burgher representation covers cities with burgher seats, where the citizenry chose at least part of the council, but no such guild seats; the rest are non-participative. Broader burgher representation stands in for his elected councils and burgher/guild for his burgher and guild representation. On my panel the specification reproduces one of his two institutional patterns (Table 5, column 2). The severe-drought main effect is positive but small and not significant, the same sign as his. In cities with broader burgher representation, severe-drought periods carry significantly less unrest, the muting he reports. In burgher/guild cities they also carry significantly less, where he finds the drought effect persisting. Both signs fit the substitution prior of Section 2, which his burgher-or-guild result runs against. To check my implementation of his method, I apply the same method as his published Table 6 (reproduced in the online appendix) to my own data.

The annual re-fit takes the same specification to the annual panel with his spatial-HAC estimator and his 100 km and 50-year cutoffs (Table 5, column 1). At annual resolution both the drought effect he finds and the muting the five-year benchmark reproduced disappear: the severe-drought main effect is essentially zero, and the broader-burgher interaction shows nothing. The drought $\times$ burgher/guild interaction is negative

Table 5: Wigton-Jones (2025) benchmark on my own data: annual and five-year periods

| | Any unrest (0/1) | |
|---|---|---|
| | Annual | Five-year periods |
| Model: | (1) | (2) |
| *Variables* | | |
| Severe drought (≤10th pctile) | -0.0001 | 0.0090 |
| | (0.0017) | (0.0077) |
| Broader burgher rep. | 0.0021 | -0.0090 |
| | (0.0021) | (0.0089) |
| Burgher/guild | 0.0051** | 0.0057 |
| | (0.0023) | (0.0107) |
| Log population | 0.0004 | 0.0022 |
| | (0.0013) | (0.0064) |
| Plague indicator | 0.0066 | 0.0024 |
| | (0.0052) | (0.0103) |
| Severe drought (≤10th pctile) × Broader burgher rep. | 0.0002 | -0.0272*** |
| | (0.0033) | (0.0095) |
| Severe drought (≤10th pctile) × Burgher/guild | -0.0062** | -0.0286** |
| | (0.0030) | (0.0130) |
| City | Yes | Yes |
| Period / year | Yes | Yes |
| *Fit statistics* | | |
| Observations | 51,632 | 10,549 |
| $R^2$ | 0.02113 | 0.08151 |
| Within $R^2$ | 0.00033 | 0.00087 |

*Custom standard-errors in parentheses*
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Column 1: annual panel (N = 51,632, 254 cities); column 2: five-year city-periods (N = 10,549, 254 cities). Both use Conley standard errors (100 km, 50 years). Severe drought is scPDSI at or below its 10th percentile of the respective panel (-2.998 in five-year periods). Reference category: non-participative government. Burgher/guild: guild seats in at least half of the period; broader burgher rep.: burgher seats without such guild seats.

and significant: severe-drought years in guild-represented cities carry *less* documented unrest, the opposite of the persistence Wigton-Jones finds there and the same pattern Section 5 finds once the years of institutional change are set aside. The choice of standard errors matters little. The spatial-HAC errors run from 8% smaller to 10% larger than city-clustered ones, and the guild interaction is the only term whose significance changes, falling to the 10% level under clustering (online appendix). Because the defensible radius for this panel is about twice the roughly 55 km cell of the Old World Drought Atlas, I also repeat the fit at 75 to 200 km (online appendix); no standard error moves more than about 1.5% from its 100 km value. Standard errors do not explain the divergence.

I then diagnose the remaining divergence one dimension at a time. I first restrict Wigton-Jones' own data and method to the 240 backbone cities his panel also covers and to the 1300–1520 window, to see whether geographic and temporal scope explain the difference. I then apply his drought and flood bins to my panel in place of my continuous measure and check the functional form, and compare the two datasets' event definitions, first through their incidence rates and then on a ladder of narrower outcomes.

Restricted to my cities and window, his data still tie drought to more years of ongoing conflict: in his

conflict-incidence column the drought coefficient stays positive and significant, and larger than published. They do not tie it to the outbreak of conflict, the measure that corresponds to my episode starts. In his conflict-start column the drought coefficient turns slightly negative, the direction of my own estimates, though too imprecisely to tell it from either zero or his published value. The rest of his pattern largely holds: 8 of 11 coefficients keep the published sign in the conflict-start column and 9 of 11 in the incidence column, and the one outlier, the guild main effect in the incidence column, is the kind of instability a much smaller sample produces. Scope cannot therefore be ruled out for outbreaks; the restriction leaves too little precision to decide.

Replacing my continuous measure with his drought and flood bins does not restore his result: the bins point the same way as the continuous measure above. Every drought bin carries a deficit of documented unrest, and the severe and extreme bins the largest, the extreme bin's slightly above the severe bin's, although no bin is significant. The bins sort each city-year by its scPDSI value, on which 0 is normal and negative values are drought: mild drought from $-1$ to $-2$, moderate from $-2$ to $-3$, severe from $-3$ to $-4$, and extreme below $-4$, with matching flood bins on the wet side, each compared with near-normal years between $-1$ and $+1$. Severe drought is thus the band just short of the extreme tail; in my annual panel it holds 8% of city-years and the extreme bin 10%. The severe-drought indicator of the two benchmarks is a different cut, the driest tenth of observations. In five-year periods it begins at about $-3.0$ and spans the severe and extreme bins together; in single years it begins at almost exactly $-4$ and so coincides with the extreme bin, though there it enters interacted with government type, so its main effect describes non-participative cities only.

Functional form does not explain the reversal. A centered quadratic in wet-dry severity, fit on an annual version of the benchmark that enters severity as a continuous measure alongside government type and on the two Wahl outcomes, adds nothing: its squared term is never distinguishable from zero, and in the guild-led unrest equation the significant linear term barely moves (online appendix). The drought bins show no interior turning point either, so over the range the data cover the reversed drought pattern is close to linear.

The two conflict measures are built differently, and the difference runs deeper than a casualty threshold. Wigton-Jones draws his events from Brecke's general historical conflict catalog, supplemented by Bradbury, and keeps an event only if he can attribute it to a specific originating city. His main source admits only conflicts with 30 or more casualties, a floor that excludes exactly the peaceful protests and unarmed refusals my catalog is built to admit. His civil category also pools two kinds of conflict, a city's population rising against its own government and a city fighting a higher authority whose power over it was only nominal, as the emperor's was over the Free Imperial Cities, and he counts city-level risings within interstate wars (Wigton-Jones, 2025, pp. 1110, 1112). My catalog's closest analog to the second kind is a refusal of homage;

Table 6: Outcome ladder: severe- and extreme-drought bins

| | | | Severe drought | | | Extreme drought | | |
|---|---|---|---|---|---|---|---|---|
| Outcome | Unrest city-years | Base rate (%) | Events | Coef. (% of base) | p | Events | Coef. (% of base) | p |
| 1. All documented unrest (paper's measure) | 345 | 0.668 | 25 | -40 | 0.11 | 29 | -47 | 0.12 |
| 2. Organized peaceful action only | 57 | 0.110 | 8 | 46 | 0.56 | 2 | -101 | 0.13 |
| 3. Any violent episode | 210 | 0.407 | 10 | -78 | 0.01 | 20 | -48 | 0.18 |
| 4. Revolution, rebellion, or coup (= violent, major severity) | 188 | 0.364 | 9 | -89 | 0.01 | 17 | -53 | 0.16 |

*Notes:*
Rungs narrow the broad documented-unrest measure, to peaceful action, which the comparator excludes, or toward his armed-conflict definition; revolution, rebellion, or coup, which here coincides with violent episodes of major severity, stands in for a casualty threshold the catalog cannot apply. Severe drought is scPDSI from -4 to -3 (4,161 city-years), extreme below -4 (5,198 of 51,632). Coef. is the bin's coefficient from the drought-bin specification as a percentage of the rung's base rate, with its city-clustered p-value; events are the rung's unrest city-years in the bin.

internal unrest against a city's own council or patriciate, the more common case in both catalogs, is closer to a like-for-like comparison. On the same 240 cities, unrest breaks out in my catalog in 0.560% of city-years, about three times his conflict-start rate of 0.183%, as a broader inclusion rule would produce, though the rates alone cannot prove it.

A direct test re-estimates the drought bins on narrower slices of my catalog (Table 6). If counting peaceful and minor episodes produced the reversal, the drought deficit should shrink and turn positive as the outcome narrows to the violent episodes both definitions count. It does not. Relative to each outcome's base rate, severe- and extreme-drought city-years carry about 40% and 47% less unrest on the broad measure, neither significant; for revolution, rebellion, or coup, which in this catalog are exactly the violent episodes of major severity and so the closest analog to a casualty threshold, about 89% and 53% less; and for any violent episode about 78% and 48%; the severe-bin deficits of both violent rungs are significant. Peaceful action alone, with only 2 events in the extreme bin, is too thin to read. Event definition therefore does not explain the divergence.

On Wigton-Jones' own data drought goes with more conflict, as published (Table 7). On mine it never goes with significantly more documented unrest. The regressions above and the Wahl guild-led unrest specification of Section 5 find the reverse, wetter years with more unrest, and the drought bins point the same way without reaching significance; the severe-drought indicators of the two benchmarks are null. The consistency of the sign carries the weight, and it is weaker than it looks, since most specifications are fit on close variants of the same panel. The diagnosis narrows the explanations without settling them. Neither the estimator nor the functional form explains the divergence, and neither does event definition, since the violent episodes both catalogs count show the deficit too. Restricted to my cities and window, his own data still tie drought to ongoing conflict, but are too imprecise on outbreaks to rule out the place and period, and thin documentation remains open too (online appendix). The grievance record bears on the question from outside the regressions: climate is a marginal presence in what the sources say the unrest was about

Table 7: Drought/climate association with conflict or unrest, across specifications

| Specification | Data | Significance | Direction |
|---|---|---|---|
| WJ Table 6, published (conflict start) | His data | 10% level | Drought associated with more conflict |
| WJ Table 6, published (conflict incidence) | His data | 1% level | Drought associated with more conflict |
| My exact replication of WJ Table 6 (appendix) | His data | Matches published | Confirms his direction |
| My WJ approximate benchmark, 5-yr | My data | Not significant | Null |
| Main specification, annual | My data | 10% level | Wetter associated with more unrest |
| My WJ benchmark re-fit, annual (severe drought) | My data | Not significant | Null |
| Drought bins, extreme-drought bin | My data | Not significant | Null |
| Climate only, annual, all cities | My data | 5% level | Wetter associated with more unrest |
| Wahl guild-transition | My data | Not significant | Null |
| Wahl guild-perpetrated unrest | My data | 5% level | Wetter associated with more unrest |

*Notes:*
WJ = Wigton-Jones (2025). Rows differ in sample, outcome, and functional form, so the table compares direction and significance, not magnitudes.

(Section 3), consistent with the argument that Flemish urban rebellion in the same broad period was the work of politicized burghers pursuing fiscal and constitutional access more than an undifferentiated hunger revolt, even where food prices were rising (Dumolyn & Haemers, 2005).

# 5 Institutions and Unrest

Representation enters the regressions next, as a covariate in its own right and as a moderator of the climate association. Section 2 set out how the council constitution entered unrest as its object, vehicle, and channel; the question here is what a city's representation went with, and when. For representation, the *Kommunalismus* channel predicts that guild and burgher seats each substitute for unrest. For guilds the effect may fade: once in power a guild could turn its organizing capacity to defending its position and close membership and citizenship to outsiders, who then had a new grievance (Ogilvie, 2004; Olson, 2003), so the guild prior is ambiguous and may change over time. Markets make coordination cheaper (Geloso & Kufenko, 2019) and anchor the prosperity that, on one reading, enriched the craftsmen who then sought power (Wahl, 2015), so their prior is positive.

Table 8 adds institutions, measured in the same year in the first four columns and the year before in the last four, and interacts wet-dry severity with each to test whether representation mutes the climate association. Guild seats go with about 60% more unrest in the same year, significant only at the 10% level, while guild seats held the year before go with about 69% *less*. Opposite signs a year apart are what Section 2's

Table 8: Climate and institutions: same-year and prior-year representation

| | Any unrest × 100 | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | Same-year institutions | | | | Prior-year institutions | | | |
| | Main | | Interaction | | Main | | Interaction | |
| Model: | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
| *Variables* | | | | | | | | |
| scPDSI | 0.0663* | | 0.0462 | | 0.0718** | | 0.0589 | |
| | (0.0343) | | (0.0345) | | (0.0342) | | (0.0361) | |
| Bottom-decile scPDSI | | -0.1370 | | -0.0131 | | -0.1603 | | -0.0211 |
| | | (0.1583) | | (0.1680) | | (0.1589) | | (0.1766) |
| Top-decile scPDSI | | 0.1488 | | 0.1423 | | 0.1431 | | 0.1567 |
| | | (0.1566) | | (0.1569) | | (0.1557) | | (0.1637) |
| Guild seats | 0.4004* | 0.4000* | 0.5316** | 0.3837* | -0.4558** | -0.4565** | -0.4177* | -0.4751** |
| | (0.2177) | (0.2179) | (0.2385) | (0.2215) | (0.2264) | (0.2264) | (0.2396) | (0.2324) |
| scPDSI × Guild seats | | | 0.1167** | | | | 0.0348 | |
| | | | (0.0504) | | | | (0.0433) | |
| Bottom-decile scPDSI × Guild seats | | | | -0.5374* | | | | -0.3001 |
| | | | | (0.3014) | | | | (0.2834) |
| Top-decile scPDSI × Guild seats | | | | 0.8944* | | | | 0.6028 |
| | | | | (0.4815) | | | | (0.4403) |
| Burgher seats | 0.0210 | 0.0178 | -0.0132 | 0.0830 | -0.0990 | -0.1024 | -0.0797 | -0.0253 |
| | (0.1990) | (0.1990) | (0.2020) | (0.2024) | (0.1903) | (0.1903) | (0.1926) | (0.1947) |
| scPDSI × Burgher seats | | | -0.0305 | | | | 0.0185 | |
| | | | (0.0378) | | | | (0.0335) | |
| Bottom-decile scPDSI × Burgher seats | | | | -0.0419 | | | | -0.2729 |
| | | | | (0.2753) | | | | (0.2559) |
| Top-decile scPDSI × Burgher seats | | | | -0.6775** | | | | -0.5247* |
| | | | | (0.2879) | | | | (0.2788) |
| Market (0/1) | 0.3140* | 0.3097* | 0.3147* | 0.3099* | 0.2645 | 0.2598 | 0.2635 | 0.2588 |
| | (0.1771) | (0.1770) | (0.1771) | (0.1768) | (0.1768) | (0.1767) | (0.1767) | (0.1765) |
| Plague indicator | 0.6553 | 0.6570 | 0.6560 | 0.6577 | 0.6803 | 0.6821 | 0.6830 | 0.6853 |
| | (0.5160) | (0.5160) | (0.5159) | (0.5159) | (0.5168) | (0.5168) | (0.5169) | (0.5171) |
| Log population | 0.0400 | 0.0385 | 0.0409 | 0.0381 | 0.0005 | -0.0010 | 0.0006 | -0.0012 |
| | (0.1338) | (0.1339) | (0.1335) | (0.1338) | (0.1342) | (0.1343) | (0.1342) | (0.1342) |
| *With guild and burgher seats* | | | | | | | | |
| scPDSI | | | 0.1323** | | | | 0.1122** | |
| | | | (0.0587) | | | | (0.0529) | |
| Bottom-decile scPDSI | | | | -0.5924* | | | | -0.5942** |
| | | | | (0.3320) | | | | (0.2965) |
| Top-decile scPDSI | | | | 0.3592 | | | | 0.2347 |
| | | | | (0.4497) | | | | (0.4287) |
| *Institution by climate variable* | | | | | | | | |
| Guild seats, median scPDSI | | | 0.4272* | | | | -0.4488** | |
| | | | (0.2209) | | | | (0.2284) | |
| Guild seats, bottom-decile scPDSI | | | | -0.1537 | | | | -0.7752** |
| | | | | (0.3278) | | | | (0.3348) |
| Guild seats, top-decile scPDSI | | | | 1.2781** | | | | 0.1277 |
| | | | | (0.5182) | | | | (0.4645) |
| Burgher seats, median scPDSI | | | 0.0141 | | | | -0.0962 | |
| | | | (0.1990) | | | | (0.1903) | |
| Burgher seats, bottom-decile scPDSI | | | | 0.0410 | | | | -0.2981 |
| | | | | (0.3317) | | | | (0.3095) |
| Burgher seats, top-decile scPDSI | | | | -0.5945* | | | | -0.5499* |
| | | | | (0.3138) | | | | (0.2988) |
| City | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Year | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| *Fit statistics* | | | | | | | | |
| Observations | 51,632 | 51,632 | 51,632 | 51,632 | 51,378 | 51,378 | 51,378 | 51,378 |
| $R^2$ | 0.02113 | 0.02107 | 0.02129 | 0.02137 | 0.02114 | 0.02107 | 0.02116 | 0.02123 |

*Conley standard errors (100 km, 50 years) in parentheses*
*Signif. codes: ***: 0.01, **: 0.05, *: 0.1*

Linear probability models on the annual city-year panel, 1300–1520, the 254 cities with institutional coverage; outcome 100 if unrest, 0 otherwise, so coefficients are in percentage points. Columns 1–4 use the same year's institutions and 5–8 the year before; even columns replace scPDSI (negative in drought) with its bottom and top deciles. In the interaction columns the climate rows refer to cities without seats, and the first block below the line gives the association where both are held.

account of representation as an outcome of unrest would produce, a point taken up below. Burgher seats, which the *Kommunalismus* channel predicts would quiet unrest, show nothing at either timing. A plague year goes with more recorded unrest in every specification, roughly doubling the usual rate, but the estimate is never significant.

The climate association keeps its sign across the table, though not its precision. The scPDSI coefficient is positive in every column but significant only in columns 1 and 5; in the interaction columns it measures the association in cities without guild or burgher seats, and where both are held it is larger and significant (rows below the line). The deciles keep their opposite signs too, about 21% less unrest in bottom-decile years and 22% more in top-decile years, though neither is significant (column 2); a quadratic term adds nothing (online appendix), and the drought × guild interaction of the Wigton-Jones re-fit in Section 4 points the same way (Wigton-Jones, 2025). With institutions and the other controls in, the association still comes from organized actors: unrest led by guilds is about 20% and unrest led by elites about 21% more frequent in a year one standard deviation wetter than usual, while burgher-led unrest shows none (online appendix). The pattern is the one Section 4 read as collective action, and it fits the guild result taken up next: better years let the best-organized actors press their claims, and sometimes win them.

Representation reshapes the climate association. Severe drought goes with less unrest, and the drop appears deepest where guilds hold seats. The guild-seats interaction with scPDSI is positive and significant, so the wetter-years association is stronger where guilds sit on the council, while the burgher-seats interaction shows nothing (column 3). The decile form locates the difference in the tails (column 4): guild seats matter at the dry end, where bottom-decile years go with less unrest than in cities without them, and burgher seats at the wet end, where top-decile years are quieter than elsewhere; the burgher interaction is significant, the guild one just short of it. In a city with both, a year one standard deviation wetter than usual goes with about 22% more unrest and a bottom-decile year with about 89% less, so representation reshapes the climate association rather than removing it. The dry end fits the collective-action reading above, with the best-organized actors quieter in the worst years; the wet end fits the *Kommunalismus* channel, in which a recognized communal voice substitutes for unrest (Blickle, 2000b), with top-decile years quieter where burghers hold seats. With last year's institutions no interaction term reaches the 5% level (columns 7 and 8), but the worst years still stand out: in a city that already held both guild and burgher seats, a bottom-decile year goes with about 90% less unrest, significantly so.

The guild main effect does not hold up. A one-year lag is enough to turn its sign, which points to the years in which representation changes rather than to the institution itself. Read from the guild side, the interaction columns show when the same-year association arises: it is concentrated in wet years, about 191% more unrest in top-decile years where guilds hold seats, significant, and nothing in bottom-decile years

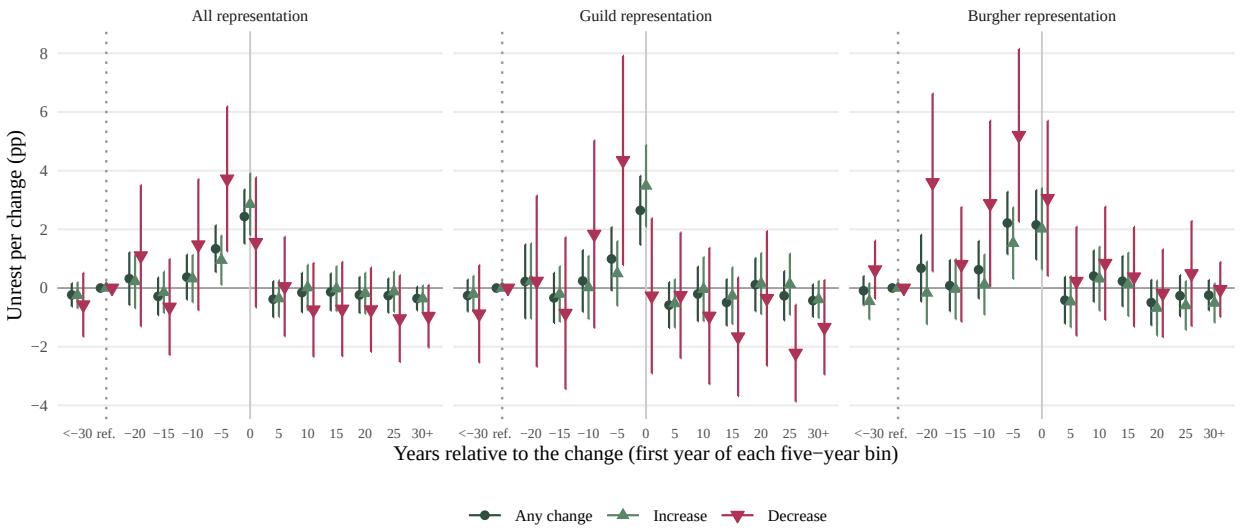



Figure 2: Recorded unrest around every change in representation. Each point is the change in unrest (percentage points) per change of that kind, in the five-year window starting at the year shown, against the decade 21 to 30 years before; the outer windows pool everything earlier or later. Changes are moves on the 0–3 scales of Table 2; all representation pools guild and burgher. Lines are 95% confidence intervals from Conley standard errors.

(column 4). The year after, the pattern inverts. Where guild government was already in place, recorded unrest all but disappears in severe drought, about 117% below the base rate and significant (column 8), with no significant change in wet years; the drop exceeds the base rate, so, as with protest, I read it as a sign rather than a size. Those are the years of the council provisioning described in Section 4, which the record cannot separate from the organizing capacity discussed there; both fit an established guild council whose worst years were quiet. They also fit the channel reading: a guild with seats could press its members' grievances inside the council, as Flemish guilds could once they sat in city government, so the actor best placed to turn a bad harvest into unrest already belonged to the order it would have challenged (Dumolyn & Haemers, 2005, p. 379; North et al., 2009; Ogilvie, 2019, p. 38). These lagged terms still rest on every city-year, change years included; Table 9 re-estimates them with the change years removed. The wet-dry gap is one of many interactions examined, so on its own it is a pattern rather than a finding, but it ties the climate and guild results together through the same events: guild-led unrest is the kind that rises in wet years, and the unrest that coincides with guilds gaining representation tended to fall in such years.

Figure 2 shows why the guild coefficients flip. It compares recorded unrest in five-year windows around each recorded change in representation with the decade 21 to 30 years before (Schmidheiny & Siegloch, 2023). In the five years from a gain in guild representation, unrest runs about 3.5 percentage points above that baseline per gain, several times the base rate; in the five years before it is barely higher, and afterwards it is

back at the baseline. Representation was often won through unrest, the concession to a credible threat that Acemoglu & Robinson (2000) model: in Dortmund the twenty-month siege of 1388–89 left the city deeply in debt, the council met it with new war taxes on sales and on wealth, and the guild rising of 1400 ended in a constitution that gave the six guilds six of the council's eighteen seats (Keyser, 1954, pp. 112–114).[8] When a revolt wins seats, its unrest falls in the first year of the new, higher representation, while the year before still records the old, lower level, so a single event yields a positive same-year coefficient and a negative lagged one. With city fixed effects the institution coefficients rest on exactly these changes, because representation is otherwise constant within a city: it changes in 285 of 52,531 city-years. Neither coefficient in Table 8 therefore measures how unrest went with the institutions a city already had.

Across all 285 recorded changes, in either direction and for both institutions, the guild pattern is not particular to guild seats or to gains. In the five years from a change in either institution, unrest runs about 2.4 percentage points above the baseline per change, and in the five years before it about 1.3; five years on it is back at its earlier level, and three decades on it shows no lasting shift in either direction. Burgher gains come out of a longer build-up than guild gains, whose unrest is concentrated in the five years from the change. Losses differ in timing: their unrest falls mostly in the five years before the loss, about 4.3 points per guild loss and 5.2 per burgher loss, with little after it, as if representation was taken back once a rising had run its course, as in the restorations of patrician rule that some guild struggles ended in (Bader & Dilcher, 1999, pp. 511–512). Losses are few (30 guild and 35 burgher), so that is the least certain part of the figure. A difference-in-differences estimator robust to staggered and heterogeneous effects gives the same answer for gains (de Chaisemartin & d'Haultfœuille, 2026). Comparing each city's first change with cities at the same level that had not yet changed, a change in guild representation goes with about 18 percentage points more unrest in the year it happens, with no significant difference in the ten years before, jointly or year by year, and essentially none on average over the ten years after; a change in burgher representation points the same way but is not significant.[9] On the sources' account, then, unrest is how representation changed, and once it had changed, the level of unrest went back to what it had been, the same answer the settled city-years give below.

Table 9 asks how unrest went with the representation a city already had, by setting the changes aside. Its first two columns repeat the main and lagged specifications on settled city-years, dropping every year

[8] In the sources for the episodes that fall in the year a city gains guild seats, the unrest comes first wherever it is described apart from the change itself. The one exception is Göttingen in 1513/14, where the council, seeking the guild masters' consent to tax increases to pay down the city's debts, first withdrew from the guilds and gave them a share in administering the city's finances; their criticism of its past conduct then turned into popular outrage that drove several councillors from the city (Postel, 1975, p. 85).

[9] The estimator follows each city's first change only, so it cannot estimate losses: a city's first move is almost always a gain, and its losses come later. Re-estimating a year-by-year event study around each city's first gain of guild seats with an interaction-weighted estimator for staggered adoption (Sun & Abraham, 2021) moves no window by more than 0.2 percentage points.

Table 9: Pre-existing institutions: settled city-years and comparisons between cities

| | Any unrest × 100 | | | | |
|---|---|---|---|---|---|
| | Main | | | Prior-year interaction | |
| | Settled | Settled, lagged | Between | scPDSI | Deciles |
| Model: | (1) | (2) | (3) | (4) | (5) |
| *Variables* | | | | | |
| scPDSI | 0.0550* | 0.0626** | 0.0538* | 0.0409 | |
| | (0.0316) | (0.0316) | (0.0294) | (0.0325) | |
| Bottom-decile scPDSI | | | | | -0.0315 |
| | | | | | (0.1603) |
| Top-decile scPDSI | | | | | 0.2336 |
| | | | | | (0.1565) |
| Guild seats | -0.0858 | -0.0928 | 0.2149* | -0.0196 | -0.0778 |
| | (0.1936) | (0.1953) | (0.1295) | (0.2113) | (0.2016) |
| scPDSI × Guild seats | | | | 0.0636 | |
| | | | | (0.0427) | |
| Bottom-decile scPDSI × Guild seats | | | | | -0.5557*** |
| | | | | | (0.2129) |
| Top-decile scPDSI × Guild seats | | | | | 0.6183 |
| | | | | | (0.4733) |
| Burgher seats | -0.0362 | -0.0363 | 0.0232 | -0.0065 | 0.0154 |
| | (0.1757) | (0.1690) | (0.0909) | (0.1752) | (0.1755) |
| scPDSI × Burgher seats | | | | 0.0285 | |
| | | | | (0.0311) | |
| Bottom-decile scPDSI × Burgher seats | | | | | -0.2146 |
| | | | | | (0.1908) |
| Top-decile scPDSI × Burgher seats | | | | | -0.3653 |
| | | | | | (0.2945) |
| Market (0/1) | 0.1869 | 0.1675 | 0.2465*** | 0.1668 | 0.1665 |
| | (0.1598) | (0.1610) | (0.0743) | (0.1607) | (0.1606) |
| Plague indicator | 0.2736 | 0.2782 | 0.4087 | 0.2823 | 0.2839 |
| | (0.4495) | (0.4495) | (0.4422) | (0.4500) | (0.4503) |
| Log population | 0.0043 | -0.0250 | 0.4193*** | -0.0243 | -0.0271 |
| | (0.1218) | (0.1213) | (0.0630) | (0.1213) | (0.1214) |
| Rye suit. | | | 0.0963 | | |
| | | | (0.2587) | | |
| *Institution by climate variable* | | | | | |
| Guild seats, median scPDSI | | | | -0.0766 | |
| | | | | (0.1982) | |
| Guild seats, bottom-decile scPDSI | | | | | -0.6335** |
| | | | | | (0.2543) |
| Guild seats, top-decile scPDSI | | | | | 0.5405 |
| | | | | | (0.4877) |
| Burgher seats, median scPDSI | | | | -0.0320 | |
| | | | | (0.1698) | |
| Burgher seats, bottom-decile scPDSI | | | | | -0.1992 |
| | | | | | (0.2307) |
| Burgher seats, top-decile scPDSI | | | | | -0.3499 |
| | | | | | (0.3105) |
| City | Yes | Yes | | Yes | Yes |
| Year | Yes | Yes | Yes | Yes | Yes |
| Region | | | Yes | | |
| *Fit statistics* | | | | | |
| Observations | 48,768 | 48,518 | 48,768 | 48,518 | 48,518 |
| $R^2$ | 0.02008 | 0.02003 | 0.01204 | 0.02012 | 0.02026 |

*Conley standard errors (100 km, 50 years) in parentheses*
*Signif. codes: ***: 0.01, **: 0.05, *: 0.1*

Linear probability models on the annual city-year panel, 1300–1520; outcome 100 if unrest, 0 otherwise, so coefficients are in percentage points. Every column drops city-years within 5 years of a change in representation (49,667 of 52,531 remain). Column 2 uses the year before; column 3 replaces city with region fixed effects and adds rye suitability; columns 4–5 repeat the prior-year interactions of Table 8, with the rows below the line giving each institution's association in a median, bottom-decile, or top-decile year.

within five years of a change in guild or burgher representation, a window that also takes in the unrest that builds before a change and any error in when it was recorded. The third compares cities with one another on the same years, with region in place of city fixed effects and rye suitability as a control, and the last two repeat the prior-year interactions of Table 8. Dropping the change years is conservative: if weak representation provoked revolts that won seats, those revolts are among the years removed, which pulls the

Table 10: Does low representation prime unrest? Representation the year before and unrest that did not change it

| | | Within cities | | Between cities | |
|---|---|---|---|---|---|
| Outcome | Episodes | Guild | Burgher | Guild | Burgher |
| All unrest | 349 | -69%** | -15% | +3% | -4% |
| No change in representation | 272 | -20% | -6% | +35% | +1% |
| Political grievance | 219 | -111%** | -8% | -1% | +5% |
| Political grievance, no change | 159 | -22% | +13% | +49% | +26% |

*Notes:*
Each cell is the coefficient on guild or burgher seats the year before, as a share of the outcome's base rate, from a linear probability model with wet-dry severity, markets, log population, and plague. Within cities: city and year fixed effects. Between cities: region and year fixed effects and rye suitability. All city-years kept. No change: representation unchanged from the year before the episode to two years after it. Conley standard errors; *** 0.01, ** 0.05, * 0.1.

settled coefficients toward zero.

Outside the years of change, representation shows no clear association with unrest. Guild seats lean negative within cities, about 16% below the base rate in the same year and 18% the year after, and positive between cities, about 40% above it; none of the three reaches the 5% level. Burgher seats show nothing in any column, so the data do not support the negative association that *Kommunalismus* predicts (Blickle, 2000b), and wet-dry severity stays positive throughout. The online appendix shows the same picture with windows of two and ten years and with the 0–3 scales. These nulls bound the association rather than rule it out. Within cities, the 95% interval for guild seats runs from about 87% below the base rate to 55% above it, and for burgher seats from about 71% below to 58% above, so the data exclude only an association close to the whole base rate; between cities the burgher interval is narrower, from about 29% below to 38% above. In the interaction columns the guild main effect is gone, which confirms that the lagged coefficient in Table 8 came from the change years, but the drought interaction is not: where guilds already held seats, recorded unrest in bottom-decile years falls to almost nothing, and the interaction itself is significant at the 1% level, while the burgher interaction in very wet years does not survive the cut. Representation, then, went with how a city met its worst years rather than with how restless it was on average, and only guild representation did.

Dropping the years of change, however, also drops the revolts that low representation may have provoked. If a recognized voice in government lowers the pressure for unrest, low representation should also raise unrest that did not itself change representation: riots, protests, failed revolts, and revolts that won something else, such as a tax concession. Table 10 tests that directly, keeping every year and changing the outcome instead. It classifies each of the 349 linked episodes by whether guild or burgher representation changed between the year before it began and two years after it ended, and by whether the catalog records a political grievance.

The within-city association runs through the episodes that changed representation. For all unrest, guild seats the year before go with about 69% less, significant; leave those episodes out and the gap shrinks to about 20%. For unrest with a political grievance it shrinks from a drop larger than its base rate, which is significant, to about 22%, which is not. Between cities, for all unrest and for political unrest, neither guild nor burgher seats go with significantly less unrest, with or without those episodes. The data therefore do not show that standing representation lowers unrest: the guild association in Table 8 comes from the episodes that changed representation, the pattern Figure 2 traced.

Holding a market goes with more unrest, as the coordination account predicts (Geloso & Kufenko, 2019), clearly so between cities, about 46% above the base rate, though not within them. The between-city estimate carries the usual caveat: market towns were larger, better connected, and better documented, and region fixed effects and log population absorb only part of that, so it likely overstates the association. It also sits beside the finding of Wahl (2015) that cities living mainly from trade saw fewer successful guild revolts, though that concerns one kind of unrest and whether it succeeded, rather than how often any unrest was recorded.

The one institutional claim Wigton-Jones tests, that representation mutes the drought-conflict link, can be read off his benchmark in Section 4 (Table 5), and the answer splits by type. Broader burgher representation mutes the severe-drought association in five-year periods, as his elected councils do, but not on the annual panel. For guild representation the sign is the opposite of his: where guilds held seats, the driest years carried the least recorded unrest, in the annual re-fit as on the settled city-years of Table 9.

The Wahl panel validated in Section 3 also supports a directional test. Wahl (2015) argues that the late medieval guild revolts had structural rather than exogenous origins: centers of proto-industry saw more successful revolts and cities living mainly from trade fewer, with the agrarian crisis and the Black Death mattering only in a wider sample of smaller cities, while pre-existing burgher representation and elections, a city's autonomy from its lord, its urban hinterland, and guild participation in neighboring cities enter as controls and spillovers. Without his replication package I replicate the argument approximately, with one reversal of framing relative to the regressions above: here guild representation is the *outcome*. I use two outcomes, a gain of guild seats and a guild-led unrest episode, and ask whether pre-existing institutions predict them. The test covers his controls and his crisis variable rather than his headline, the proto-industry and trade contrast, which no linked source records. The spillover term counts neighboring cities within 100 km with a prior guild revolt, a proxy for his share of neighbors with guild participation within 150 km.

Table 11 reports the replication. Pre-existing burgher representation predicts a later gain of guild seats: such gains are rare (77 across 41,270 city-years), and each step up the burgher-representation scale adds about 3 times that base rate to the yearly chance. With a later guild-led unrest episode as the outcome (156

Table 11: Wahl (2015) guild-revolt replication: institutions and climate as predictors

| | Gain of guild seats (0/1) | Guild-led unrest (0/1) |
|---|---|---|
| | Guild transition | Guild-led unrest |
| Model: | (1) | (2) |
| *Variables* | | |
| Burgher rep. (0–3) | 0.0064*** | 0.0010 |
| | (0.0020) | (0.0009) |
| Autonomy proxy | 0.0012 | $-1.72 \times 10^{-5}$ |
| | (0.0022) | (0.0013) |
| scPDSI | $1.49 \times 10^{-5}$ | 0.0005** |
| | (0.0002) | (0.0002) |
| Log population | $9.18 \times 10^{-5}$ | -0.0006 |
| | (0.0008) | (0.0008) |
| Spatial lag (100km) | 0.0012 | 0.0017 |
| | (0.0012) | (0.0011) |
| City | Yes | Yes |
| Year | Yes | Yes |
| *Fit statistics* | | |
| Observations | 41,270 | 53,078 |
| $R^2$ | 0.02940 | 0.01570 |
| Within $R^2$ | 0.00266 | 0.00025 |

*Clustered (City) standard-errors in parentheses*
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

scPDSI is negative in drought. The autonomy proxy is 1 where the territorial ruler holds no governing role in the city. Outcomes: a first gain of guild seats, after which the city leaves the risk set (N = 41,270 city-years, 254 cities, 77 events); and guild-perpetrated unrest (N = 53,078 city-years, 262 cities, 156 events). Annual panel; city and year fixed effects.

across 53,078 city-years), it shows no significant association. Wahl carries pre-existing burgher representation only as an unreported control, so this tests a mechanism he holds fixed rather than his headline; what it shows is that one group's representation predicted the next group's entry. Wet-dry severity shows nothing in the transition equation and is significant and positive, meaning wetter, in the guild-led unrest equation, the opposite of Wahl's agricultural-crisis mechanism and the same reversed sign as the climate tables of Section 4. The autonomy proxy is null in both equations, plausibly because it is coarse. The spillover term is null for both outcomes, so his neighborhood mechanism does not reappear at this resolution.

Across Section 4 and this section, three patterns stand out. Wetter years go with more recorded unrest, carried by organized actors rather than by unrest in general; changes in representation coincide with unrest, so the representation coefficients a panel regression returns reflect the change years; and where guilds already held seats, the driest years were the quietest. The comparisons with Wigton-Jones and Wahl leave each of them in place. The institutional coefficients also describe only part of the backbone. The regression sample (51,632 of 67,626 city-years, 254 cities) is smaller than the full panel because representation is missing wherever no in-window or carried-forward entry exists, following the documentation-survival pattern described in Section 3, so the institutional results speak only to cities with a recorded constitutional history. The online appendix reports the same specifications with coups as the outcome, without the market control,

by half-century, and by form of unrest.

# 6 Conclusion

This paper builds a link the literature on the Holy Roman Empire lacked: a geolocated, event-level record of popular unrest matched, city by city and year by year, to a comparably resolved record of who governed, across 306 cities from 1300 to 1520. Two demonstrations show what the link supports. The first asks whether the closest published comparator's finding, that drought raises unrest and that representation through elected councils mutes the effect (Wigton-Jones, 2025), reproduces in a panel built for the Empire. The drought finding does not, and the muting reproduces only in part, for burgher representation in five-year periods; the staged diagnosis built to explain the drought divergence rules out the estimator and the functional form (Section 4), cannot rule out geographic and temporal scope for outbreaks of conflict, finds that event definition does not explain the divergence, and leaves thin documentation open.

Thin documentation is harder to rule out. If drought-era episodes were failing to reach the secondary literature, the deficit should be deepest where the record is thinnest; one documentation cut points that way, one points the other, and the size cut shows no contrast; no subsample is significant (online appendix).

What the diagnosis does establish concerns event definition. My catalog records outbreaks of unrest about three times as often as Wigton-Jones' casualty-thresholded conflict measure records outbreaks of conflict, as a broader event criterion on my side would produce. Yet the difference in definition does not explain the reversal. Violent episodes, which both definitions would admit, show a drought-year deficit at least as deep as the catalog as a whole, significantly so in severe-drought years, and no narrower outcome reverses the sign (Section 4). The peaceful episodes that most separate the two definitions are too few to read in the drought bins; the scarcity of protest in the driest years rests instead on the type split of Table 4, where it fits the collective-action reading (Section 4).

For the climate-conflict literature, the implication is narrower than a claim that drought was unrelated to unrest in the pre-industrial Empire. The negative coefficient is estimated on documented episodes in 306 cities. A casualty-thresholded conflict measure and a broader measure that includes peaceful protest and refusals of homage alongside armed revolt are related constructs, and a finding that holds for one need not hold for the other even within the same population and period, but here the reversal holds for the violent episodes both would count. Whether it reflects a feature of the late medieval Empire absent from a longer and broader sample, or the way the Empire's unrest reached the record, remains open. The breakdown by grievance and conflict type (online appendix) sharpens the question without settling it: economic grievance, which includes every named natural shock and is the category a subsistence-shock account would most

directly implicate, is rare and, where it can be tested, also goes with wetter rather than drier years, and the grievance record is thin on subsistence language generally (Section 3). Climate may have mattered through channels the grievance record does not name, such as the room a good year gives organized actors to press their claims, or the association may be too weak to carry much weight.

The second demonstration concerns institutional timing. Unrest clusters in the years in which guilds gained representation, and those years produce the guild associations a standard panel regression returns, positive in the same year and negative the year after; away from them, neither guild nor burgher representation shows a clear association with unrest, though a moderate within-city effect cannot be ruled out (Section 5). Pre-existing burgher representation predicts later gains of guild seats, though not guild-led unrest, so it went with the rare successful entries rather than with guild agitation in general (Section 5). Neither result is the specific claim Wigton-Jones tests, that representation mutes the drought-conflict relationship. For guild representation that claim returns the opposite sign: where guilds already sat on the council, the driest years carry the least recorded unrest, a result that survives once the years of institutional change are set aside (Section 5). Representation, then, went with how a city met its worst years rather than with how much unrest it had. A dataset resolved enough to keep these claims apart is itself part of the contribution. Markets go with more unrest, clearly so when cities are compared with one another, in line with the coordination account, though that comparison carries the usual cross-sectional caveats (Section 5).

Two features of the construction bound what the diagnosis can claim. The first is the possible selectivity of Franz (1977), one of the catalog's main sources: a survey organized around the Peasants' War is likely to understate the smaller, locally resolved disputes that never fed into its story. The second is the study population, restricted to cities large enough to enter the population data by 1500, a criterion plausibly correlated with the institutional stability the representation measures capture. If institutionally fragile cities were also more likely to decline, miss the population threshold, or lose documentary continuity, the panel underrepresents the least robust cities, so its estimates describe the more durable ones and may understate a stabilizing association.

Together, the reversed drought association, the dominance of political and fiscal grievances in the complaint record (Section 3), and the deliberately broad event criterion fit a reading in which documented urban unrest in this setting more often worked as an instrument of institutional bargaining than as a subsistence-driven eruption, closer, in the terms of Section 2, to unrest over the council as its object, carried by organized vehicles, and to the moral-economy account than to a deprivation account keyed to the shock itself. The grievance record cannot adjudicate that reading against a competing one. Chroniclers who recorded political and fiscal complaints may simply have been less likely to record a subsistence one, so the thin subsistence record (Section 3) is consistent with either reading: bargaining dominated, or subsistence grievance went

unrecorded.

If the bargaining reading holds, several things should follow. First, because urban communes had standing councils and guild structures that could absorb a grievance inside an existing forum, while rural populations more often had to build the vehicle for collective action from scratch, extending the catalog to rural episodes through a grid-cell design should show a weaker institutional association and a stronger climate association in the rural subsample. A rural extension that reproduced the urban pattern would undercut that account. Second, if unrest functioned as political negotiation under weak institutionalized voice, the extension of judicial access after 1495 through the *Reichskammergericht*, and above all the 1526 settlement that referred disputes between lords and subjects to the courts, should displace it, with the documented-unrest rate falling in a series continued past 1520 as litigation substitutes for revolt. Schulze finds that shift for the early modern Empire: after the Peasants' War, peasant conflict was increasingly channeled into litigation before the territorial courts and the *Reichskammergericht*, a "tendential juridification of social conflict" that he dates from the 1526 Reichsabschied and that subjects pursued so persistently that it set the German lands apart from France, England, and Hungary (Schulze, 1980, pp. 76–77, 141).

Third, set against matched English and French series over 1350–1400, where a tight post-plague cluster of wage, tax, and war-finance grievances produced compressed revolt waves, the German series should lack a synchronized spike and remain dominated by political grievances, consistent with an agrarian depression that deepened over a century and struck regions very unevenly (Abel, 2013, pp. 66, 81, 86). A synchronized German spike would be evidence against it. Fourth, the drought-by-guild interaction and the organized-actor pattern were found in these data rather than specified in advance, so they should be checked on data not used to find them. In a catalog extended beyond this window or region, cities whose guilds already held seats should again record quieter severe-drought years, and any wet-year association should again be carried by rebellions and by guild- and elite-led unrest rather than by the riots and peasant risings a subsistence account would implicate.

Two smaller extensions follow. First, the panel's ordinal scales could be deepened: actual seat counts in place of the 0–3 bands and the seats indicator, and the formal procedures for removing officeholders, would sharpen mechanism tests the current coding can only gesture at, and the guild results should be re-run under alternative rulings on reserved seats, electoral colleges, and merchant elites. Second, a target-based cut reported in the online appendix identifies eleven geocoded episodes of unrest directed at least in part at Jewish communities: too few to test formally, but enough to show the data support the question Anderson, Johnson, and Koyama (2017) ask at pan-European scale, whether weather shocks predict this specific form of targeted violence. Institutional heterogeneity across the Empire predicted the intensity of Black Death-era anti-Jewish violence (Finley & Koyama, 2018), an institution-by-persecution question a panel matched at

the city-year rather than territory level could in principle speak to at finer resolution. Strasbourg's 1349 pogrom is a cautionary example for how institutions enter such a regression: the city was not under patrician rule, since the conflict of 1332 had ended the lineages' hold on the council, which from 1333 seated twenty-five guild representatives alongside fourteen burghers and eight nobles under a newly created Ammeister (Naegle & Solórzano Telechea, 2014). On 9 February 1349, after butchers demanded that the craftsmen receive a share of the money the Jewish community had paid and the presiding magistrate, Peter Swarber, refused, the craftsmen assembled in arms before the cathedral, forced him and two other councillors to resign, and installed a new council, which four days later resolved to burn the city's Jews (Winkler, 2017). A participatory channel and persecution could thus sit on the same side of a single conflict, the channel serving as the vehicle rather than the brake a cross-city panel regression would implicitly assume.

The debates this paper started from, over why late medieval communities revolted and whether admitting guilds and burghers to government calmed or fed that unrest, have long rested on a few well-archived cities. Linking an event-level record of unrest to an annual record of who governed does not settle them. It does move them onto evidence that can be counted, dated, and checked city by city, and it shows that at least one pattern drawn from a broader European sample does not carry over intact to the Empire.

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